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PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 4

INTEGRATION

SUGGESTED TIME: 9 PERIODS

IN THIS CHAPTER

This chapter develops the definite integral as the limit of Riemann sums and applies it to areas, volumes of revolution, centroids and centres of mass. It closes with numerical integration by the trapezoid rule and the mean value theorem for integrals, essential tools for the GCE Advanced Level examination.

- By the end of this chapter I can express a definite integral as the limit of a Riemann sum and interpret it as a signed area.
- By the end of this chapter I can use definite integrals to calculate the area between a curve and the x-axis, including regions below the axis.
- By the end of this chapter I can calculate the area between two curves and areas bounded by the y-axis.
- By the end of this chapter I can calculate volumes of solids of revolution about the x-axis and about the y-axis.
- By the end of this chapter I can use integration to locate centroids of plane regions and centres of mass of simple laminas and solids.
- By the end of this chapter I can approximate a definite integral with the trapezoid rule and estimate the accuracy of the approximation.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Section 1: The Definite Integral as a Limit of Riemann Sums
2. Section 2: Areas Bounded by the x-axis and Between Curves
3. Section 3: Areas Bounded by the y-axis
4. Section 4: Volumes of Solids of Revolution
5. Section 5: Centroids and Centres of Mass by Integration
6. Section 6: Trapezoid Rule and the Mean Value Theorem for Integrals
7. Integration activity
8. Methods and worked examples
9. Exercises
10. Solutions to the exercises
11. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can express a definite integral as the limit of a Riemann sum and interpret it as a signed area.
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- By the end of this chapter I can use integration to locate centroids of plane regions and centres of mass of simple laminas and solids.
- By the end of this chapter I can approximate a definite integral with the trapezoid rule and estimate the accuracy of the approximation.
- By the end of this chapter I can state and apply the mean value theorem for integrals and compute the mean value of a function on an interval.
- By the end of this chapter I can model and solve authentic GCE-style problems combining several of these skills.

PREREQUISITES

- Indefinite integration and the Fundamental Theorem of Calculus (Lower Sixth: antiderivatives, standard integrals, integration by substitution and by parts).
- Evaluation of definite integrals with correct handling of limits.
- Curve sketching of polynomials, rational, trigonometric, exponential and logarithmic functions.
- Solving simultaneous equations to find intersection points of curves.
- Sigma notation and limits of sums (sequences and series, Lower Sixth).

1 SECTION 1: THE DEFINITE INTEGRAL AS A LIMIT OF RIEMANN SUMS

A water engineer in Bamenda must estimate the volume of a reservoir whose cross-section is curved, while a cocoa cooperative in Kumba wants to know the average daily price of a bag of cocoa over the season. A road contractor between Douala and Yaoundé needs the exact area of an irregular plot bounded by a curving river. None of these shapes is a rectangle, a triangle or a circle, so ordinary geometry fails. Integration, understood as the limit of a sum of infinitely many tiny pieces, gives the exact answer — and the trapezoid rule gives a quick estimate when the function is only known from measurements.

1.1 Why We Need a New Idea: Area Under a Curve

In earlier years you learned to compute areas of polygons, circles and sectors. But what if the boundary is a curve like $y = x^2$ or $y = \sin x$? There is no elementary formula for the area under such a curve between $x = a$ and $x = b$. The brilliant idea — due to Riemann and others — is to exhaust the region with rectangles whose total area we can sum, then let the number of rectangles grow without bound while their width shrinks to zero. The limiting value, if it exists, is defined to be the definite integral of the function over $[a, b]$.

1.2 Partitioning the Interval and Rectangular Approximations

Let f be a function defined on a closed interval $[a, b]$. A partition P of $[a, b]$ is a finite set of points $\{x_0, x_1, \dots, x_n\}$ such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

The subintervals are $[x_{i-1}, x_i]$ and their widths are $\Delta x_i = x_i - x_{i-1}$. The mesh (or norm) of the partition, denoted $\|P\|$, is the length of the largest subinterval: $\|P\| = \max_{1 \le i \le n} \Delta x_i$.

For a regular partition into n equal subintervals,

$$\Delta x = \frac{b - a}{n}, \qquad x_i = a + i\Delta x \quad (i = 0, 1, \dots, n).$$

On each subinterval $[x_{i-1}, x_i]$ we choose a sample point $x_i^* \in [x_{i-1}, x_i]$ and build a rectangle of height $f(x_i^*)$ and width Δx_i. The sum of the areas of these rectangles is a Riemann sum:

$$S(P) = \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i.$$

For a regular partition with $\Delta x_i = \Delta x$, this simplifies to

$$S_n = \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

Common choices for x_i^* in a regular partition:

- **Left endpoint:** $x_i^* = x_{i-1}$ (left Riemann sum).
- **Right endpoint:** $x_i^* = x_i$ (right Riemann sum).

- **Midpoint:** $x_i^* = \frac{x_{i-1}+x_i}{2}$ (midpoint Riemann sum).

If f is increasing, the left sum underestimates the area (lower sum) and the right sum overestimates it (upper sum); for a decreasing function the roles are reversed.



Left Riemann sums for $y = x^2$ on $[0,1]$ with $n = 4$ and $n = 10$. As n increases the rectangles better approximate the area under the curve.

1.3 Sigma Notation and Riemann Sums

The Riemann sum is compactly written with sigma notation:

$$S_n = \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

For a regular partition $\Delta x = \frac{b-a}{n}$ and right endpoints $x_i^* = a + i\Delta x$, this becomes

$$S_n = \sum_{i=1}^{n} f\left(a + i\,\frac{b-a}{n}\right)\frac{b-a}{n}.$$

EXEMPLE **Riemann sum for $f(x) = x^2$ on $[0,1]$**

Take $a = 0$, $b = 1$, so $\Delta x = \frac{1}{n}$ and $x_i^* = \frac{i}{n}$. Then

$$S_n = \sum_{i=1}^{n} \left(\frac{i}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3}\sum_{i=1}^{n} i^2.$$

Using the well-known formula $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$, we obtain

$$S_n = \frac{1}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

EXEMPLE **Riemann sum for $f(x) = \frac{1}{x}$ on $[1,2]$**

Here $a = 1$, $b = 2$, $\Delta x = \frac{1}{n}$, $x_i^* = 1 + \frac{i}{n}$. The right Riemann sum is

$$S_n = \sum_{i=1}^{n} \frac{1}{1+\frac{i}{n}}\cdot\frac{1}{n} = \sum_{i=1}^{n} \frac{1}{n+i}.$$

This sum does not simplify to a closed form using elementary formulas, but its limit as $n \to \infty$ exists and equals $\ln 2$.

METHOD **Evaluating an integral from first principles**

To compute $\int_a^b f(x)\,dx$ directly from the definition:

1. Write $\Delta x = \frac{b-a}{n}$ and choose a convenient sample point (usually right endpoint $x_i = a + i\Delta x$).
2. Form the Riemann sum $S_n = \sum_{i=1}^{n} f(x_i)\,\Delta x$.
3. Simplify the sum using known formulas: $\sum_{i=1}^{n} 1 = n$, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$.
4. Take the limit $n \to \infty$ (i.e. $\Delta x \to 0$). The result is the value of the definite integral.

1.4 The Definite Integral as a Limit

DEFINITION **Definite integral as a limit of Riemann sums**

Let f be defined on $[a, b]$. If the limit

$$\lim_{\|P\| \to 0} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i$$

exists and is the same for every choice of partitions P and sample points $x_i^* \in [x_{i-1}, x_i]$ as the mesh $\|P\|$ tends to zero, then f is said to be integrable on $[a, b]$ and the limit is called the definite integral of f from a to b, denoted

$$\int_a^b f(x)\,dx.$$

For a regular partition this is equivalent to $\lim_{n \to \infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x$.

PROPERTY **Integrability of continuous functions**

If f is continuous on the closed interval $[a, b]$, then f is integrable on $[a, b]$. (This theorem is examinable; you may be asked to state it.)

WORKED EXAMPLE 1 **Evaluate $\int_0^1 x^2\,dx$ from the definition**

We use the right Riemann sum computed earlier:

$$S_n = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

Taking the limit as $n \to \infty$:

$$\int_0^1 x^2\,dx = \lim_{n\to\infty} S_n = \frac{1}{3}.$$

SOLUTION

Check: Using the Fundamental Theorem of Calculus, $\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$. The two methods agree.

1.5 Connection with the Fundamental Theorem of Calculus

The limit-of-sums definition and the antiderivative method are two faces of the same coin. The **Fundamental Theorem of Calculus** (FTC) states that if f is continuous on $[a, b]$ and F is any antiderivative of f (i.e. $F' = f$), then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Thus the definite integral, defined as a limit of Riemann sums, can be evaluated in practice by finding an antiderivative. The FTC also guarantees that the limit of Riemann sums exists for every continuous function, reinforcing the integrability theorem above.

1.6 Properties of the Definite Integral

All the following properties follow directly from the definition as a limit of sums (linearity of limits, additivity of partitions, order preservation).

PROPERTY **Linearity, additivity and order properties of the definite integral**

Let f and g be integrable on $[a, b]$ and c a constant.

1. Linearity: $\int_a^b \big(cf(x) + g(x)\big)\,dx = c\int_a^b f(x)\,dx + \int_a^b g(x)\,dx.$
2. Additivity over intervals: If $a < c < b$, then $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$
3. Reversal of limits: $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$. In particular $\int_a^a f(x)\,dx = 0.$
4. Non-negativity: If $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x)\,dx \geq 0.$

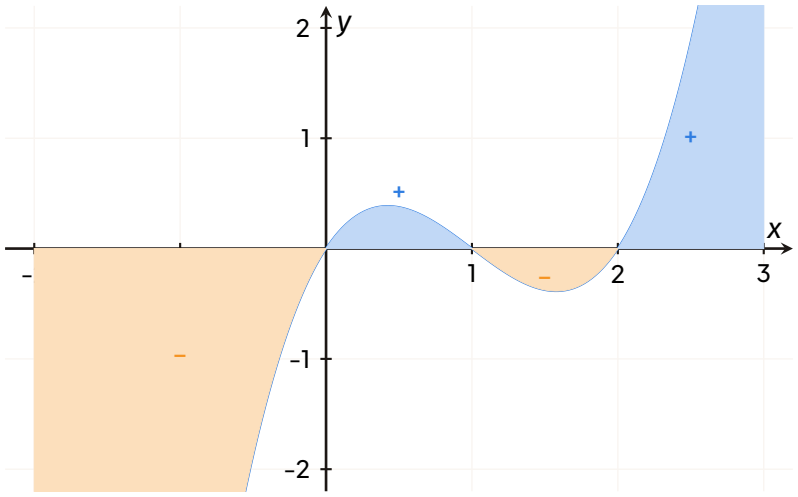
5. Comparison: If $f(x) \le g(x)$ on $[a, b]$, then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$.

1.7 Signed Area Interpretation

The definite integral computes signed area: regions where $f(x) > 0$ contribute positively, regions where $f(x) < 0$ contribute negatively. The integral is not the total geometric area unless the function does not change sign.

WARNING **Signed area vs. geometric area**

A negative value of $\int_a^b f(x)\,dx$ does not mean "negative surface". It means the net signed area is negative because the part below the x-axis outweighs the part above. To find the actual geometric area between the curve and the x-axis, you must split the interval at the zeros of f and integrate $|f(x)|$.



Signed area for $y = x(x-1)(x-2)$ on $[-2, 3]$. Blue regions (above x-axis) contribute positively; orange regions (below x-axis) contribute negatively. The definite integral is the algebraic sum.

1.8 Worked Example: Evaluating an Integral from First Principles

WORKED EXAMPLE 2 **Compute $\int_1^3 (2x+1)\,dx$ using the definition of the definite integral**

Step 1: Partition $[1, 3]$ into n equal subintervals.

$$\Delta x = \frac{3-1}{n} = \frac{2}{n}, \qquad x_i = 1 + i\Delta x = 1 + \frac{2i}{n}.$$

Step 2: Right Riemann sum:

$$S_n = \sum_{i=1}^{n} f(x_i)\,\Delta x = \sum_{i=1}^{n} \left(2\left(1 + \frac{2i}{n}\right) + 1\right)\frac{2}{n} = \sum_{i=1}^{n} \left(3 + \frac{4i}{n}\right)\frac{2}{n}.$$

Step 3: Distribute and separate sums:

$$S_n = \frac{6}{n}\sum_{i=1}^{n} 1 + \frac{8}{n^2}\sum_{i=1}^{n} i = \frac{6}{n}\cdot n + \frac{8}{n^2}\cdot\frac{n(n+1)}{2} = 6 + 4\frac{n+1}{n} = 6 + 4 + \frac{4}{n} = 10 + \frac{4}{n}.$$

Step 4: Take the limit $n \to \infty$:

$$\int_1^3 (2x+1)\,dx = \lim_{n\to\infty}\left(10 + \frac{4}{n}\right) = 10.$$

SOLUTION

Verification with FTC: An antiderivative of $2x + 1$ is $x^2 + x$.

$$\left[x^2 + x\right]_1^3 = (9 + 3) - (1 + 1) = 12 - 2 = 10.$$

Both methods give the same result.

1.9 Exercises

EXERCISE 1 **Practice with Riemann sums**

1. Write the right Riemann sum for $f(x) = x^3$ on $[0, 2]$ with n subintervals. Simplify using $\sum i^3$ and find $\int_0^2 x^3\,dx$ by taking the limit.
2. For $f(x) = \sqrt{x}$ on $[0, 4]$, set up the left Riemann sum with n subintervals. (You do not need to evaluate the limit.)
3. Use the definition to evaluate $\int_0^2 (3x^2 - 2x)\,dx$.
4. Explain why $\int_{-1}^{1} x^3\,dx = 0$ without computing an antiderivative, using the signed area interpretation.

SOLUTION **Exercise 1**

$\Delta x = \frac{2}{n}$, $x_i = \frac{2i}{n}$. $S_n = \sum_{i=1}^{n}\left(\frac{2i}{n}\right)^3\frac{2}{n} = \frac{16}{n^4}\sum_{i=1}^{n} i^3 = \frac{16}{n^4}\left[\frac{n(n+1)}{2}\right]^2 = \frac{16}{n^4}\cdot\frac{n^2(n+1)^2}{4} = 4\frac{(n+1)^2}{n^2} = 4\left(1 + \frac{1}{n}\right)^2$. Limit $n \to \infty$ gives 4. Check: $\int_0^2 x^3\,dx = \left[\frac{x^4}{4}\right]_0^2 = \frac{16}{4} = 4$.

SOLUTION **Exercise 2**

$\Delta x = \frac{4}{n}$, left endpoints $x_{i-1} = \frac{4(i-1)}{n}$ for $i = 1, \dots, n$. Left Riemann sum: $S_n = \sum_{i=1}^{n} \sqrt{\frac{4(i-1)}{n}} \cdot \frac{4}{n} = \frac{8}{n\sqrt{n}} \sum_{i=1}^{n} \sqrt{i-1} = \frac{8}{n^{3/2}} \sum_{k=0}^{n-1} \sqrt{k}$. (No elementary closed form; the limit as $n \to \infty$ gives $\int_0^4 \sqrt{x}\,dx = \frac{16}{3}$.)

SOLUTION **Exercise 3**

$\Delta x = \frac{2}{n}$, $x_i = \frac{2i}{n}$. $S_n = \sum_{i=1}^{n} \left[3\left(\frac{2i}{n}\right)^2 - 2\left(\frac{2i}{n}\right)\right]\frac{2}{n} = \sum_{i=1}^{n} \left(\frac{24i^2}{n^3} - \frac{8i}{n^2}\right) = \frac{24}{n^3}\frac{n(n+1)(2n+1)}{6} - \frac{8}{n^2}\frac{n(n+1)}{2} = 4\frac{(n+1)(2n+1)}{n^2} - 4\frac{n+1}{n} = 4\left(2 + \frac{3}{n} + \frac{1}{n^2}\right) - 4\left(1 + \frac{1}{n}\right) = 8 + \frac{12}{n} + \frac{4}{n^2} - 4 - \frac{4}{n} = 4 + \frac{8}{n} + \frac{4}{n^2}$. Limit = 4. Check: $\int_0^2 (3x^2 - 2x)\,dx = [x^3 - x^2]_0^2 = 8 - 4 = 4$.

SOLUTION **Exercise 4**

The function $f(x) = x^3$ is odd: $f(-x) = -f(x)$. On the symmetric interval $[-1, 1]$, the region above the x-axis on $[0, 1]$ is exactly mirrored below the x-axis on $[-1, 0]$. The signed areas cancel exactly, so the definite integral is zero.

2 SECTION 2: AREAS BOUNDED BY THE X-AXIS AND BETWEEN CURVES

In **Section 1** we built the definite integral from below: we sliced a region into thin rectangles, summed their areas, and passed to the limit to obtain the Riemann integral $\int_a^b f(x)\,dx$. We now harvest the fruit of that construction. Because the definite integral is, by its very definition, a limit of areas of rectangles, it is the natural tool for computing areas exactly. In this section we answer three questions of increasing subtlety: How do we find the area between a curve and the x-axis? What happens when the curve dips below the axis? And how do we find the area trapped between two curves? These questions appear every year at the GCE Advanced Level, usually carrying between 6 and 10 marks, so mastering them is essential.

2.1 Area between $y = f(x) \geq 0$ and the x-axis

Suppose f is continuous and **non-negative** on $[a, b]$. The region bounded by the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is exactly the region whose area the Riemann sums of Section 1 approximate. Passing to the limit gives the fundamental formula.

PROPERTY **Area under a positive curve**

If f is continuous and $f(x) \geq 0$ for all $x \in [a, b]$, then the area of the region bounded by $y = f(x)$, the x-axis, $x = a$ and $x = b$ is

$$A = \int_a^b f(x)\,dx.$$

EXEMPLE **A first routine calculation**

Find the area bounded by $y = x^2 + 1$, the x-axis and the lines $x = 0$ and $x = 2$.
Since $x^2 + 1 > 0$ everywhere, we integrate directly:

$$A = \int_0^2 (x^2 + 1)\,dx = \left[\frac{x^3}{3} + x\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3} \text{ square units.}$$

EXEMPLE **A trigonometric area**

Find the area under one arch of $y = \sin x$.
One arch lies above the x-axis on $[0, \pi]$, since $\sin x \geq 0$ there. Hence

$$A = \int_0^{\pi} \sin x\,dx = [-\cos x]_0^{\pi} = -\cos\pi + \cos 0 = 1 + 1 = 2 \text{ square units.}$$

Notice the elegance: the area under a whole arch of the sine curve is exactly 2.

2.2 Regions below the x-axis: the absolute value

Here is the crucial subtlety. The definite integral computes a signed quantity: where $f(x) < 0$, the products $f(x_i)\,\Delta x$ in the Riemann sum are negative, so the integral counts that part of the region negatively. But an area, as a physical quantity, is always positive. If a farmer near Buea measures a plot of land, the answer cannot be -40 square metres!

DID YOU KNOW? **Area and signed integrals**

The word "quadrature" (from the Latin quadratura, "making square") was the historical term for finding areas. Ancient Greek mathematicians like Archimedes used the method of exhaustion — a geometric ancestor of Riemann sums — to find the area under a parabola. The modern integral notation $\int$ is an elongated 'S' for summa (sum), introduced by Leibniz. The distinction between signed area (the integral) and geometric area (the absolute integral) is the reason we must split at zeros.

PROPERTY **Area when the curve crosses the axis**

If f is continuous on $[a, b]$, the total area between the curve $y = f(x)$ and the x-axis on $[a, b]$ is

$$A = \int_a^b |f(x)|\,dx.$$

Concretely, if $f(x) \le 0$ on $[c, d]$, the area of that part is $A = -\int_c^d f(x)\,dx = \int_c^d \big(-f(x)\big)\,dx$.

This means that when f changes sign inside $[a, b]$, we must **split** the integral at every zero of f, integrate each piece separately, and take the absolute value of each result before adding.

METHOD **Four-step strategy for area with the x-axis**

1. Sketch. Draw the curve $y = f(x)$ on $[a, b]$ (a rough sketch is enough: sign and zeros matter, not beauty).
2. Zeros. Solve $f(x) = 0$ to find where the curve meets the x-axis inside $[a, b]$.
3. Split. Break $[a, b]$ into sub-intervals on which f keeps a constant sign.
4. Absolute value. Integrate on each sub-interval, take the absolute value of each partial integral, and add: $A = \sum \left|\int f(x)\,dx\right|$.

WARNING **The cancellation trap**

Integrating across a zero of f without splitting makes the positive and negative parts cancel, giving an answer that is too small — sometimes even zero or negative, which is absurd for an area. For example,

$$\int_0^{2\pi} \sin x\,dx = [-\cos x]_0^{2\pi} = -1 + 1 = 0,$$

yet the true area between $y = \sin x$ and the x-axis on $[0, 2\pi]$ is $2 + 2 = 4$ square units. A zero or negative "area" in an examination is a signal that you forgot to split.

WORKED EXAMPLE 3 **Area of a cubic crossing the axis**

Find the total area enclosed between the curve $y = x(x - 1)(x - 2)$ and the x-axis.

SOLUTION

Step 1 — Sketch and zeros. The cubic factorises, so its zeros are $x = 0, 1, 2$. A sign study gives: $y > 0$ on $(0, 1)$ and $y < 0$ on $(1, 2)$ (the curve starts negative for $x < 0$, crosses up at 0, down at 1, up at 2).

Step 2 — Expand. $y = x(x-1)(x-2) = x^3 - 3x^2 + 2x$, with antiderivative $F(x) = \frac{x^4}{4} - x^3 + x^2$.

Step 3 — Split and integrate.

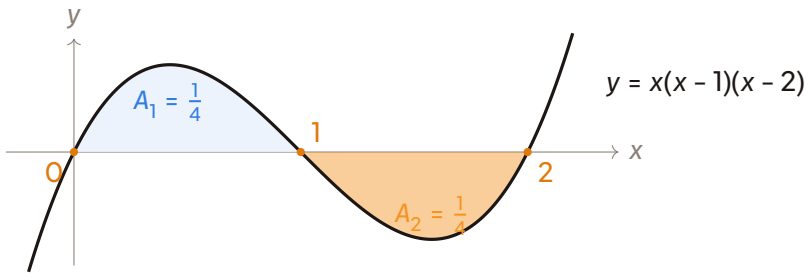
$$\int_0^1 y\,dx = F(1) - F(0) = \left(\tfrac{1}{4} - 1 + 1\right) - 0 = \tfrac{1}{4},$$

$$\int_1^2 y\,dx = F(2) - F(1) = (4 - 8 + 4) - \tfrac{1}{4} = -\tfrac{1}{4}.$$

Step 4 — Absolute values and total.

$$A = \left|\tfrac{1}{4}\right| + \left|-\tfrac{1}{4}\right| = \tfrac{1}{2} \text{ square units.}$$

Had we integrated from 0 to 2 in one go, we would have obtained $F(2) - F(0) = 0$: the two equal lobes would have cancelled completely!



The cubic $y = x(x-1)(x-2)$ crosses the x-axis three times. The two lobes must be integrated separately: $A = |A_1| + |A_2|$.

2.3 Area between two curves

Now suppose two continuous curves $y = f(x)$ and $y = g(x)$ satisfy $f(x) \geq g(x)$ on $[a, b]$. Slice the region between them into thin vertical strips of width Δx: each strip is approximately a rectangle of height $f(x) - g(x)$, hence of area $\big(f(x) - g(x)\big)\Delta x$. Summing and passing to the limit (exactly the Riemann-sum argument of Section 1) yields the following property.

PROPERTY **Top curve minus bottom curve**

If f and g are continuous with $f(x) \geq g(x)$ for all $x \in [a, b]$, then the area of the region be-

tween the two curves from $x = a$ to $x = b$ is

$$A = \int_a^b [f(x) - g(x)]\,dx \qquad \text{(top - bottom)}.$$

This formula remains valid even if one or both curves lie below the x-axis, because the difference of heights is what matters.

Very often the limits a and b are not given: they are the x-coordinates of the **intersection points** of the two curves, found by solving $f(x) = g(x)$.

WORKED EXAMPLE 4 **Area between a parabola and an inverted parabola**

Find the area of the region enclosed by $y = x^2$ and $y = 2 - x^2$.

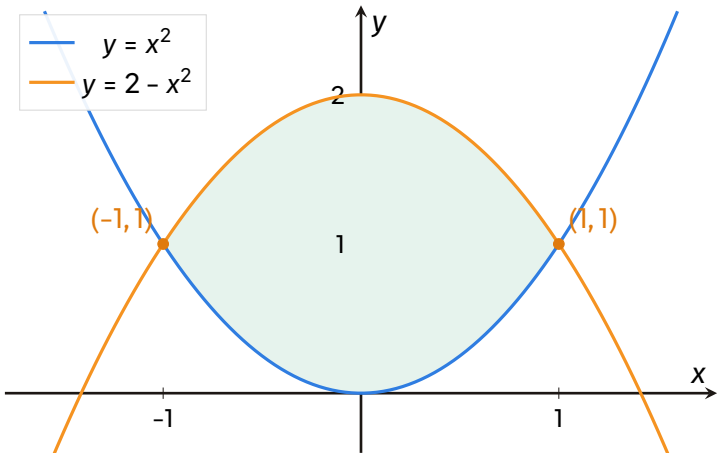
SOLUTION

Intersections. $x^2 = 2 - x^2 \iff 2x^2 = 2 \iff x = \pm 1$. The intersection points are $(-1, 1)$ and $(1, 1)$.
Which curve is on top? Test $x = 0$: $2 - 0^2 = 2 > 0 = 0^2$, so $y = 2 - x^2$ is the top curve on $[-1, 1]$.
Integrate top minus bottom.

$$A = \int_{-1}^{1} [(2 - x^2) - x^2]\,dx = \int_{-1}^{1} (2 - 2x^2)\,dx = 2\int_{-1}^{1} (1 - x^2)\,dx$$

$$= 2\left[x - \frac{x^3}{3}\right]_{-1}^{1} = 2\left[\left(1 - \tfrac{1}{3}\right) - \left(-1 + \tfrac{1}{3}\right)\right] = 2 \times \frac{4}{3} = \frac{8}{3}.$$

The enclosed area is $\frac{8}{3}$ square units. (Quick check by symmetry: $A = 2\int_0^1 (2 - 2x^2)\,dx = 4\left[x - \frac{x^3}{3}\right]_0^1 = 4 \times \frac{2}{3} = \frac{8}{3}$. ✓)



The region enclosed by $y = x^2$ and $y = 2 - x^2$, with intersection points $(\pm 1, 1)$ marked. Area $= \int_{-1}^{1}[(2 - x^2) - x^2]\,dx = \frac{8}{3}$.

2.4 Curves that cross inside the interval

When the two curves cross inside the interval of interest, the identity of the "top" curve changes. We must then split at each intersection and integrate $|f(x) - g(x)|$ piece by piece — exactly the same philosophy as for the x-axis.

WORKED EXAMPLE 5 **Two crossing curves**

Find the total area of the regions enclosed between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$.

SOLUTION

Intersections. $\sin x = \cos x \iff \tan x = 1 \iff x = \frac{\pi}{4}$ in $[0, \pi]$.

Which is on top? At $x = 0$: $\cos 0 = 1 > \sin 0 = 0$, so cosine is on top on $\left[0, \frac{\pi}{4}\right]$. At $x = \frac{\pi}{2}$: $\sin \frac{\pi}{2} = 1 > \cos \frac{\pi}{2} = 0$, so sine is on top on $\left[\frac{\pi}{4}, \pi\right]$.

Split and integrate.

$$A_1 = \int_0^{\pi/4} (\cos x - \sin x)\,dx = [\sin x + \cos x]_0^{\pi/4} = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - 1 = \sqrt{2} - 1,$$

$$A_2 = \int_{\pi/4}^{\pi} (\sin x - \cos x)\,dx = [-\cos x - \sin x]_{\pi/4}^{\pi} = (1 - 0) - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) = 1 + \sqrt{2}.$$

Total. $A = A_1 + A_2 = (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2}$ square units.

WARNING **Verify which curve is on top — on every sub-interval**

Never assume the same curve stays on top throughout. After finding each intersection point, test a convenient value of x in each sub-interval (or use your sketch) to decide the order. Writing $\int_a^b (f-g)\,dx$ when g overtakes f halfway produces cancellation and a wrong, too-small answer — the same trap as crossing the x-axis.

2.5 GCE favourites: exponentials, logarithms and mixed curves

The GCE Board loves combining areas with the exponential and logarithmic functions, because they test both your integration techniques and your curve-sketching. Recall the key antiderivatives:

$$\int e^x\,dx = e^x + C, \qquad \int \frac{1}{x}\,dx = \ln|x| + C, \qquad \int \ln x\,dx = x\ln x - x + C.$$

WORKED EXAMPLE 6 **Area involving the exponential**

Find the area bounded by $y = e^x$, the line $y = e$, and the y-axis.

SOLUTION

Sketch and intersections. $y = e^x$ meets $y = e$ where $e^x = e$, i.e. $x = 1$. The y-axis is $x = 0$. On $[0, 1]$, the line $y = e$ is above the curve (since $e^x \le e$ for $x \le 1$).
Integrate top minus bottom.

$$A = \int_0^1 (e - e^x)\,dx = [ex - e^x]_0^1 = (e - e) - (0 - 1) = 1 \text{ square unit.}$$

A remarkably clean answer — typical of well-chosen GCE questions.

EXEMPLE **Area under a hyperbola branch**

Find the area under $y = \frac{1}{x}$ from $x = 1$ to $x = 2$.

$$A = \int_1^2 \frac{dx}{x} = [\ln x]_1^2 = \ln 2 - \ln 1 = \ln 2 \text{ square units.}$$

Leave the answer as $\ln 2$: it is exact, and examiners reward exactness.

2.6 Examination technique

METHOD **Presenting an area question at the GCE**

1. Sketch first. Even a rough diagram earns method marks and protects you from sign errors. Mark the intersection points and shade the required region.
2. Justify the limits. Show the equation you solved (e.g. $x^2 = 2 - x^2$) to obtain a and b; do not pull limits out of thin air.
3. State top minus bottom (or $|f|$) explicitly before integrating, with a one-line justification ("on $[0, 1]$, $e \geq e^x$").
4. Split at every zero or crossing, and take absolute values of the pieces.
5. Give the exact answer and finish with the words "square units".

WARNING **Exact answers unless told otherwise**

Leave your final answers in exact form — surds, π, e, $\ln 2$ — unless the question explicitly requests a decimal approximation. Writing $A \approx 2.667$ instead of $A = \frac{8}{3}$ can cost the final accuracy mark. If a decimal is requested, give it to the required accuracy after stating the exact value.

EXERCISE 2 **Practice**

1. Find the area bounded by $y = 4 - x^2$ and the x-axis.
2. Find the total area between $y = x^3$ and the x-axis from $x = -1$ to $x = 2$.
3. Find the area enclosed by $y = x^2$ and $y = x$.
4. Find the area enclosed by $y = x^2 - 2x$ and $y = x$.
5. Find the area bounded by $y = \cos x$, the x-axis, $x = 0$ and $x = \pi$ (careful!).
6. Find the area of the region bounded by $y = \ln x$, the x-axis and the line $x = e$.

SOLUTION **Exercises 1–6**

1. Zeros: $x = \pm 2$; curve above axis on $[-2, 2]$. $A = \int_{-2}^{2} (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{32}{3}$.

2. Split at $x = 0$: $\int_{-1}^{0} x^3\,dx = -\frac{1}{4}$, $\int_{0}^{2} x^3\,dx = 4$. Total $A = \frac{1}{4} + 4 = \frac{17}{4}$.

3. Intersections: $x^2 = x \Rightarrow x = 0, 1$; line on top. $A = \int_0^1 (x - x^2)\,dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.
4. Intersections: $x^2 - 2x = x \Rightarrow x(x-3) = 0 \Rightarrow x = 0, 3$; line on top. $A = \int_0^3 (3x - x^2)\,dx = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \frac{27}{2} - 9 = \frac{9}{2}$.
5. $\cos x \geq 0$ on $[0, \frac{\pi}{2}]$, $\cos x \leq 0$ on $[\frac{\pi}{2}, \pi]$. $A = \int_0^{\pi/2} \cos x\,dx - \int_{\pi/2}^{\pi} \cos x\,dx = 1 - (-1) = 2$.
6. $\ln x \geq 0$ for $x \geq 1$, so the region runs from $x = 1$ to $x = e$: $A = \int_1^e \ln x\,dx = [x \ln x - x]_1^e = (e - e) - (0 - 1) = 1$ square unit.

KEY POINTS **Key points of Section 2**

- $f \geq 0$ on $[a, b]$: area $= \int_a^b f(x)\,dx$.
- In general: area $= \int_a^b |f(x)|\,dx$ — split at zeros, take absolute values.
- Between two curves: $A = \int_a^b (\text{top} - \text{bottom})\,dx$; find a, b from intersections; split if the curves cross.
- Always sketch, always justify the limits, always finish with an exact answer in square units.

In the next section we rotate our point of view by ninety degrees: instead of integrating with respect to x, we shall compute areas bounded by the y-axis, integrating with respect to y — a small change of perspective that greatly simplifies many GCE problems.

3 SECTION 3: AREAS BOUNDED BY THE Y-AXIS

In Section 2 we learned to compute areas by slicing a region into thin vertical strips of width dx and height $f(x)$, leading to the formula $A = \int_a^b f(x)\,dx$. But nature does not always present regions in the most convenient orientation. Imagine a tailor in Buea cutting a piece of cloth bounded on the left by the vertical edge of the table (the y-axis) and on the right by a curved seam: it is far more natural to measure the cloth horizontally, layer by layer from bottom to top. Mathematically this means slicing the region into horizontal strips of thickness dy and length x, and integrating with respect to y. This section develops that idea rigorously, shows how to rewrite $y = f(x)$ as $x = g(y)$ safely, and teaches you how to choose the variable of integration wisely — a skill that the GCE examiners test regularly.

3.1 Area Between a Curve $x = g(y)$, the y-axis and the Lines $y = c$, $y = d$

EXPERIMENT **Horizontal strips and Riemann sums**

Consider the region bounded by the curve $x = g(y)$ (with $g(y) \geq 0$), the y-axis, and the horizontal lines $y = c$ and $y = d$ ($c < d$). Partition $[c, d]$ into n subintervals of equal width $\Delta y = \dfrac{d - c}{n}$, and in each subinterval pick a sample point y_k^*. The horizontal strip at height y_k^* is approximately a rectangle of height Δy and length $g(y_k^*)$, hence of area $g(y_k^*)\,\Delta y$. Summing,

$$A \approx \sum_{k=1}^{n} g(y_k^*)\,\Delta y.$$

This is a Riemann sum of the function g on $[c, d]$. As $n \to \infty$ the sum converges to the definite integral $\displaystyle\int_c^d g(y)\,dy$, exactly as in Section 1 — only the roles of x and y are interchanged.

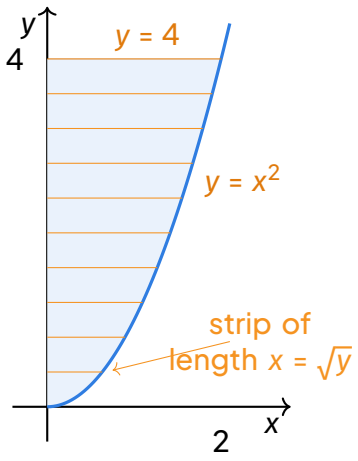
PROPERTY **Area bounded by the y-axis**

Let g be continuous on $[c, d]$ with $g(y) \geq 0$ for all $y \in [c, d]$. The area of the region bounded by the curve $x = g(y)$, the y-axis and the lines $y = c$ and $y = d$ is

$$A = \int_c^d g(y)\,dy = \int_c^d x\,dy.$$

If $g(y) \leq 0$ on $[c, d]$ (curve to the left of the y-axis), the area is $A = \displaystyle\int_c^d \left|g(y)\right| dy = -\int_c^d g(y)\,dy$.

The justification is the Riemann-sum argument above; you may be asked to reproduce it briefly, so keep the picture of a horizontal strip of area $x\,\Delta y$ in mind.



Horizontal strips for the region bounded by $y = x^2$, the y-axis and $y = 4$: each strip has length $x = \sqrt{y}$ and thickness dy.

EXEMPLE **First computation with respect to y**

Find the area bounded by the curve $y = x^2$, the y-axis and the lines $y = 0$ and $y = 4$.
Solution. On the region of interest $x \geq 0$, so $y = x^2$ gives $x = \sqrt{y}$ (positive branch). Then

$$A = \int_0^4 \sqrt{y}\,dy = \left[\frac{2}{3}y^{3/2}\right]_0^4 = \frac{2}{3}(8 - 0) = \frac{16}{3} \text{ square units.}$$

3.2 Rewriting $y = f(x)$ as $x = g(y)$: Choosing the Correct Branch

To integrate with respect to y we must express x as a function of y. This is not always possible globally: the equation $y = f(x)$ may have several solutions x for a given y, and we must select the branch that actually borders our region.

WARNING **Check the branch on the whole interval**

When solving $y = f(x)$ for x, check the sign and branch of x on the whole interval $[c, d]$.

- $y = x^2$ with $x \geq 0$ gives $x = \sqrt{y}$; with $x \leq 0$ it gives $x = -\sqrt{y}$. Using $x = \sqrt{y}$ for a region lying to the left of the y-axis gives a wrong (even negative-looking) answer.
- If the curve crosses the y-axis inside $[c, d]$, split the integral: $A = \int_c^d |g(y)|\,dy$ must be broken where g changes sign.
- Some relations cannot be solved nicely for x (e.g. $y = x^3 + x$); then integration with respect to x is preferable.

WORKED EXAMPLE 7 **Region to the left of the y-axis**

Find the area bounded by the curve $y = x^2$, the y-axis, and the lines $y = 1$ and $y = 4$, the region lying to the left of the y-axis.

SOLUTION

Solution. Here $x \leq 0$, so the correct branch is $x = -\sqrt{y}$, which is negative on $[1, 4]$. The area is

$$A = \int_1^4 \left|-\sqrt{y}\right| dy = \int_1^4 \sqrt{y}\,dy = \left[\frac{2}{3}y^{3/2}\right]_1^4 = \frac{2}{3}(8 - 1) = \frac{14}{3} \text{ square units.}$$

Note the answer is positive, as an area must be.

3.3 The Complement Method: Rectangle Minus $\int f(x)\,dx$

There is often a second route to the same area. The region bounded by $y = f(x)$, the y-axis and

the lines $y = 0$, $y = f(b)$ (with f increasing on $[0, b]$, $f(0) = 0$) fits inside the rectangle $[0, b] \times [0, f(b)]$, and

$$\underbrace{\int_0^b f(x)\,dx}_{\text{under the curve}} + \underbrace{\int_0^{f(b)} g(y)\,dy}_{\text{left of the curve}} = b\,f(b),$$

so that

$$A = b\,f(b) - \int_0^b f(x)\,dx.$$

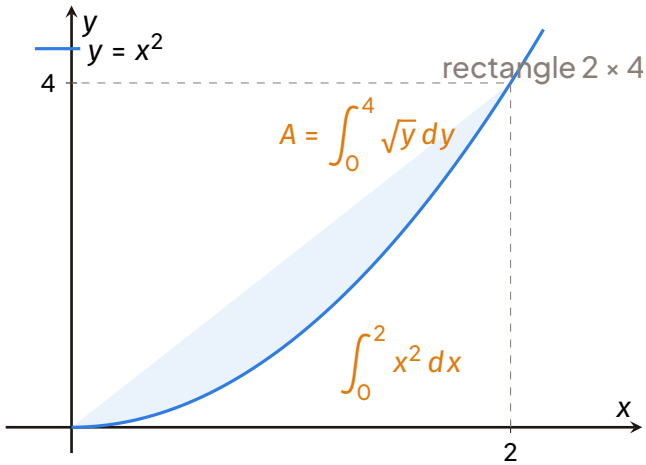
This identity is the geometric heart of the integration by parts formula, and it gives an excellent way to check an answer.

EXEMPLE **The same area, two ways**

For the region bounded by $y = x^2$, the y-axis, $y = 0$ and $y = 4$: the bounding rectangle has width 2 and height 4, hence area 8. The area under the curve from $x = 0$ to $x = 2$ is

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

Therefore $A = 8 - \frac{8}{3} = \frac{16}{3}$, confirming the direct computation $\int_0^4 \sqrt{y}\,dy = \frac{16}{3}$.



The shaded area equals both $\int_0^4 \sqrt{y}\,dy$ and $2 \times 4 - \int_0^2 x^2\,dx = \frac{16}{3}$.

DID YOU KNOW? **Integration by parts is geometry**

The formula $\int_0^b f(x)\,dx + \int_0^{f(b)} f^{-1}(y)\,dy = b\,f(b)$ for a strictly increasing function f with $f(0) = 0$ is exactly the statement that the two regions fill the rectangle. Substituting $y = f(x)$ in

the second integral yields $\int u\,dv = uv - \int v\,du$: integration by parts is nothing more than splitting a rectangle!

3.4 Area Between Two Curves Expressed as Functions of y

If a region is bounded on the right by $x = g_2(y)$ and on the left by $x = g_1(y)$, with $g_2(y) \geq g_1(y)$ for all $y \in [c, d]$, then each horizontal strip has length $g_2(y) - g_1(y)$, and

$$A = \int_c^d \left[g_2(y) - g_1(y)\right] dy.$$

This is the exact analogue of the "top curve minus bottom curve" rule of Section 2, with "right minus left" replacing "top minus bottom". Note that the y-axis itself is simply the curve $x = 0$, so the earlier formula is the special case $g_1 \equiv 0$.

WORKED EXAMPLE 8 **Integrating with respect to y is simpler**

Find the area of the region bounded by the parabola $y^2 = 4x$ and the line $y = x - 3$.

SOLUTION

Solution. Intersection points. Substituting $x = y+3$ into $y^2 = 4x$: $y^2 = 4y+12$, i.e. $y^2-4y-12 = 0$, so $(y - 6)(y + 2) = 0$ and $y = -2$ or $y = 6$.
Choice of variable. With respect to y, the region is bounded on the right by the line $x = y+3$ and on the left by the parabola $x = \dfrac{y^2}{4}$ on the whole interval $[-2, 6]$ — one single integral. (With respect to x we would need two integrals, because the lower boundary changes from the lower branch of the parabola to the line at $x = 1$.)
Computation.

$$A = \int_{-2}^{6} \left[(y + 3) - \frac{y^2}{4}\right] dy = \left[\frac{y^2}{2} + 3y - \frac{y^3}{12}\right]_{-2}^{6}$$
$$= (18 + 18 - 18) - \left(2 - 6 + \frac{2}{3}\right) = 18 - \left(-\frac{10}{3}\right) = \frac{64}{3} \text{ square units.}$$

3.5 Deciding the Variable of Integration: Mixed GCE Problems

GCE questions typically describe "the region bounded by the curve … , the x-axis / y-axis and the lines … " and expect you to set up and evaluate the integral yourself. The decision procedure below will serve you throughout the chapter (areas now, volumes and centroids later).

METHOD **Choosing between integration with respect to x and to y**

1. Sketch the region: draw the curve(s), the axis and the given lines; mark intersection points by solving the relevant equations.
2. Draw one test strip, vertical then horizontal. Ask: does each strip of a given orientation have both ends on the same two boundaries throughout the region?
3. If vertical strips work with a single pair of boundaries, integrate with respect to x: $A = \int_a^b [\text{top} - \text{bottom}]\,dx$.
4. If horizontal strips work with a single pair of boundaries, integrate with respect to y: $A = \int_c^d [\text{right} - \text{left}]\,dy$.
5. If neither works with a single integral, choose the orientation that needs the fewest pieces, or use the complement method (rectangle minus a simpler integral).
6. Check that your integrand is non-negative on each piece, and sanity-check the numerical answer against an estimate from the sketch.

WARNING **Exam tip: the quick sketch saves you**

A 20-second sketch with one horizontal strip drawn on it prevents the two most frequent errors seen by examiners: (i) using the wrong branch when solving for x in terms of y, and (ii) integrating $g(y)$ when $|g(y)|$ is required. Marks are also awarded for a correct, clearly labelled diagram — never skip it.

WORKED EXAMPLE 9 **A mixed GCE-style problem**

The curve $y = x^3$ meets the line $y = 8$ at the point P. Find the area of the finite region bounded by the curve, the y-axis and the line $y = 8$.

SOLUTION

Solution. Intersection: $x^3 = 8 \Rightarrow x = 2$, so $P = (2, 8)$.
Method 1 (with respect to y). Solving for x: $x = y^{1/3} \geq 0$ on $[0, 8]$. Hence

$$A = \int_0^8 y^{1/3}\,dy = \left[\frac{3}{4}y^{4/3}\right]_0^8 = \frac{3}{4}(16) = 12 \text{ square units.}$$

Method 2 (complement). The rectangle $[0, 2] \times [0, 8]$ has area 16, and

$$\int_0^2 x^3\,dx = \left[\frac{x^4}{4}\right]_0^2 = 4, \qquad A = 16 - 4 = 12.$$

Both methods agree: $A = 12$ square units.

EXERCISE 3 **Practice**

1. Find the area bounded by the curve $x = y^2 + 1$, the y-axis and the lines $y = 0$, $y = 2$.
2. Find the area bounded by $y = \sqrt{x}$, the y-axis and the line $y = 3$, (a) integrating with respect to y, (b) by the complement method.
3. Find the area of the region enclosed by $x = y^2$ and $x = 2 - y^2$.
4. The region bounded by $y = 2x - x^2$ and the x-axis is given. Explain why integrating with respect to y requires two pieces, and compute the area with respect to x instead.

SOLUTION **Exercise 1**

1. $x = y^2 + 1 \geq 0$ on $[0, 2]$: $A = \int_0^2 (y^2 + 1)\,dy = \left[\frac{y^3}{3} + y\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3}$.
2. (a) $x = y^2$, $A = \int_0^3 y^2\,dy = 9$. (b) Rectangle $9 \times 3 = 27$ minus $\int_0^9 \sqrt{x}\,dx = \frac{2}{3}(27) = 18$: $A = 27 - 18 = 9$.
3. Intersections: $y^2 = 2 - y^2 \Rightarrow y = \pm 1$. Right curve $x = 2 - y^2$, left curve $x = y^2$: $A = \int_{-1}^1 (2 - 2y^2)\,dy = \left[2y - \frac{2y^3}{3}\right]_{-1}^1 = \frac{8}{3}$.
4. Solving $y = 2x - x^2$ gives $x = 1 \pm \sqrt{1 - y}$: the region is bounded on the left and right by the parabola, so horizontal strips run from $1 - \sqrt{1 - y}$ to $1 + \sqrt{1 - y}$ — doable but awkward. With respect to x: $A = \int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

KEY POINTS **Key points of Section 3**

- Area between $x = g(y) \geq 0$, the y-axis, $y = c$, $y = d$: $A = \int_c^d g(y)\,dy$.
- If the curve lies to the left of the y-axis, integrate $|g(y)|$; split the integral at any sign change.
- Solving $y = f(x)$ for x: choose the branch matching the region (e.g. $x = \sqrt{y}$ for $x \geq 0$).
- Complement method: $A = b\,f(b) - \int_0^b f(x)\,dx$ — a powerful check.

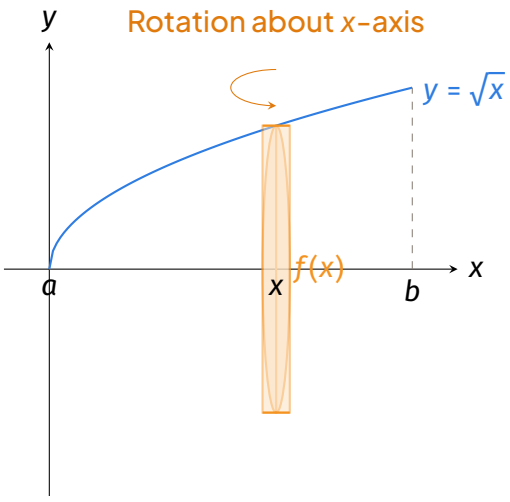
- Between two curves of y: $A = \int_c^d [\text{right} - \text{left}]\,dy$.
- Always sketch first; let the strips choose the variable of integration for you.

4 SECTION 4: VOLUMES OF SOLIDS OF REVOLUTION

In the previous section we learned how to compute areas bounded by curves and the coordinate axes. We now extend this idea to three dimensions: when a plane region is rotated about a line (an axis) in its plane, it generates a solid of revolution. The volume of such a solid can be obtained by slicing it into infinitesimally thin discs or washers and summing their volumes — a direct application of the definite integral as a limit of Riemann sums.

4.1 The Disc Method: Rotation about the x-axis

Consider a continuous function $y = f(x) \geq 0$ on the interval $[a, b]$. The region bounded by the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is rotated completely about the x-axis. This generates a solid whose cross-sections perpendicular to the x-axis are circular discs.



Region under $y = \sqrt{x}$ on $[0, 4]$ rotated about the x-axis. A typical disc of radius $f(x)$ and thickness dx is highlighted.

A typical disc at position x has radius $f(x)$ and thickness dx. Its volume is $dV = \pi[f(x)]^2\,dx$. Summing these discs from $x = a$ to $x = b$ and taking the limit gives the definite integral.

PROPERTY **Volume of Revolution about the x-axis (Disc Method)**

If f is continuous and non-negative on $[a, b]$, the volume of the solid generated by rotating the region under $y = f(x)$ about the x-axis is

$$V_x = \pi \int_a^b [f(x)]^2 \, dx.$$

This formula is derived directly from the limit of a Riemann sum of disc volumes and is examinable in the GCE.

KEY POINTS **Key Steps for the Disc Method about the x-axis**

- Sketch the region and identify the axis of rotation.
- The radius of a typical disc is the y-value of the curve: $R = f(x)$.
- The thickness is dx; the limits of integration are the x-bounds of the region.
- Write $V = \pi \int_a^b [f(x)]^2 \, dx$ and evaluate.
- Always check: the integrand is $[f(x)]^2$, not $f(x^2)$, and the factor π is essential.

EXEMPLE **Standard Solids: Cone, Sphere, Cylinder**

Verify the well-known volume formulas using the disc method.

1. Cone: Rotate the line $y = \frac{r}{h}x$ on $[0, h]$ about the x-axis.

$$V = \pi \int_0^h \left(\frac{r}{h}x\right)^2 dx = \pi \frac{r^2}{h^2}\left[\frac{x^3}{3}\right]_0^h = \frac{1}{3}\pi r^2 h.$$

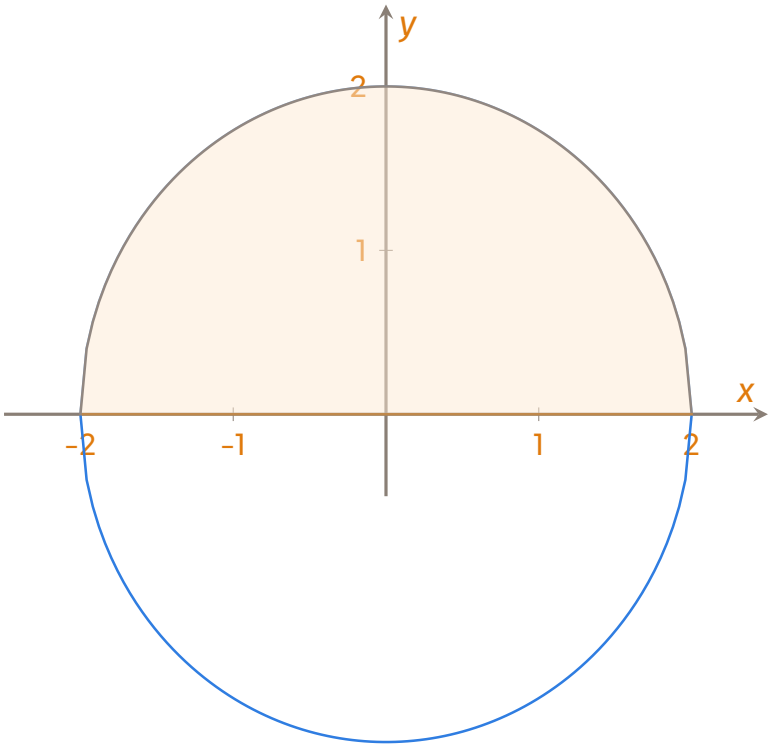
2. Sphere: Rotate the semicircle $y = \sqrt{r^2 - x^2}$ on $[-r, r]$ about the x-axis.

$$V = \pi \int_{-r}^{r} (r^2 - x^2) \, dx = \pi \left[r^2 x - \frac{x^3}{3}\right]_{-r}^{r} = \frac{4}{3}\pi r^3.$$

3. Cylinder: Rotate the horizontal line $y = r$ on $[0, h]$ about the x-axis.

$$V = \pi \int_0^h r^2 \, dx = \pi r^2 h.$$

Semicircle $y = \sqrt{r^2 - x^2}$ rotated about x-axis → Sphere



The semicircle $y = \sqrt{4 - x^2}$ (radius $r = 2$) rotated about the x-axis generates a sphere of volume $\frac{4}{3}\pi r^3$.

WORKED EXAMPLE 10 **Volume of a Paraboloid**

Find the volume of the solid generated by rotating the region bounded by $y = x^2$, the x-axis, and $x = 2$ about the x-axis.

SOLUTION

Solution: The curve is $y = x^2 \geq 0$ on $[0, 2]$. Using the disc method:

$$V = \pi \int_0^2 (x^2)^2 \, dx = \pi \int_0^2 x^4 \, dx = \pi \left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5} \text{ units}^3.$$

4.2 The Disc Method: Rotation about the y-axis

If the region is bounded by a curve expressed as $x = g(y)$ (with $g(y) \geq 0$) on $[c, d]$ and rotated about the y-axis, the discs are perpendicular to the y-axis. The radius is now $g(y)$ and the thickness is dy.

PROPERTY **Volume of Revolution about the y-axis (Disc Method)**

If g is continuous and non-negative on $[c, d]$, the volume of the solid generated by rotating the region bounded by $x = g(y)$, the y-axis, and $y = c$, $y = d$ about the y-axis is

$$V_y = \pi \int_c^d [g(y)]^2 \, dy.$$

WARNING **Radii as Functions of y**

When rotating about the y-axis, the radius of each disc must be expressed as a function of y. If the curve is given as $y = f(x)$, you must either:

- solve for $x = g(y)$ explicitly, or
- use the washer method directly with respect to y (see below).

Forgetting to change variable is a frequent error in GCE examinations.

EXEMPLE **Volume about the y-axis**

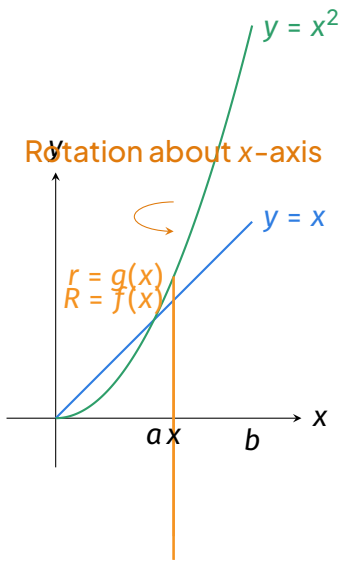
Find the volume when the region bounded by $y = x^2$, $y = 4$, and the y-axis is rotated about the y-axis.

Solution: The curve is $x = \sqrt{y}$ on $[0, 4]$. Radius $R = \sqrt{y}$.

$$V = \pi \int_0^4 (\sqrt{y})^2 \, dy = \pi \int_0^4 y \, dy = \pi \left[\frac{y^2}{2} \right]_0^4 = 8\pi \text{ units}^3.$$

4.3 The Washer Method: Region Between Two Curves

When the rotated region is bounded by two curves $y = f(x)$ (upper) and $y = g(x)$ (lower) with $f(x) \geq g(x) \geq 0$ on $[a, b]$, the cross-section is a washer (a disc with a hole). The outer radius is $R = f(x)$, the inner radius is $r = g(x)$.



Washer cross-section for the region between $y = x$ (outer) and $y = x^2$ (inner) rotated about the x-axis.

PROPERTY **Washer Method about the x-axis**

If f and g are continuous on $[a, b]$ with $f(x) \geq g(x) \geq 0$, the volume generated by rotating the region between them about the x-axis is

$$V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2\right) \mathrm{d}x.$$

METHOD **Disc vs Washer — Step by Step**

1. Sketch the region and the axis of rotation.
2. Identify the outer radius R (distance from axis to outer curve) and the inner radius r (distance from axis to inner curve). If the region touches the axis, $r = 0$ (disc method).
3. Determine the variable of integration (x for rotation about x-axis, y for rotation about y-axis).
4. Write the integral: $V = \pi \int (R^2 - r^2)\, \mathrm{d}(\text{variable})$.
5. Evaluate and give the answer in exact form (multiples of π) unless a decimal approximation is requested.

WORKED EXAMPLE 11 **Washer Method Example**

Find the volume of the solid obtained by rotating the region bounded by $y = x$ and $y = x^2$ about the x-axis.

SOLUTION

Solution: The curves intersect at $x = 0$ and $x = 1$. On $[0, 1]$, $x \geq x^2 \geq 0$. Outer radius $R = x$, inner radius $r = x^2$.

$$V = \pi \int_0^1 \left(x^2 - (x^2)^2\right) dx = \pi \int_0^1 (x^2 - x^4)\, dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \pi\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{2\pi}{15} \text{ units}^3.$$

4.4 Volumes about the y-axis from $y = f(x)$: Washer Method

Often the curve is given as $y = f(x)$ but the rotation is about the y-axis. The standard approach is to use the washer method with respect to y. For a region bounded by $y = f(x)$, $y = 0$, $x = a$, $x = b$ (with $0 \leq a < b$ and f increasing), the cross-sections perpendicular to the y-axis are washers. The outer radius is the constant b (the right boundary), and the inner radius is a for $0 \leq y \leq f(a)$, and $g(y)$ (the inverse of f) for $f(a) \leq y \leq f(b)$.

PROPERTY **Washer Method about the y-axis for a Region under $y = f(x)$**

Let f be continuous and increasing on $[a, b]$ with $f(a) \geq 0$. The volume of the solid generated by rotating the region bounded by $y = f(x)$, $y = 0$, $x = a$, $x = b$ about the y-axis is

$$V = \pi \int_0^{f(a)} (b^2 - a^2)\, dy + \pi \int_{f(a)}^{f(b)} \left(b^2 - [g(y)]^2\right) dy,$$

where $x = g(y)$ is the inverse function of f. If $a = 0$, this simplifies to

$$V = \pi \int_0^{f(b)} \left(b^2 - [g(y)]^2\right) dy.$$

EXEMPLE **Washer Method about the y-axis**

Find the volume of the solid generated by rotating the region bounded by $y = x^2$, $y = 0$, $x = 1$, and $x = 2$ about the y-axis.

Solution: The region is bounded by $y = x^2$, the x-axis, and the vertical lines $x = 1$, $x = 2$. Rotating about the y-axis. The curve $y = x^2$ gives $x = \sqrt{y}$ (inverse function $g(y) = \sqrt{y}$). y-limits: at $x = 1$, $y = 1$; at $x = 2$, $y = 4$. For $y \in [0, 1]$, the cross-section is a washer with outer radius $R = 2$ and inner radius $r = 1$. For $y \in [1, 4]$, the cross-section is a washer with outer

radius $R = 2$ and inner radius $r = \sqrt{y}$.

$$V = \pi \int_0^1 (2^2 - 1^2)\,dy + \pi \int_1^4 \left(2^2 - (\sqrt{y})^2\right) dy = \pi \int_0^1 3\,dy + \pi \int_1^4 (4 - y)\,dy$$

$$= \pi[3y]_0^1 + \pi\left[4y - \frac{y^2}{2}\right]_1^4 = 3\pi + \pi((16 - 8) - (4 - 0.5)) = 3\pi + \pi(8 - 3.5) = 3\pi + 4.5\pi = 7.5\pi = \frac{15\pi}{2}\ \text{units}^3.$$

4.5 Complementary Volume Method (Cylinder Minus Inner Solid)

An alternative technique, often quicker when the inverse function is simple, is to consider the solid as a cylinder with a cavity removed.

PROPERTY **Complementary Volume Method**

For the region bounded by $y = f(x)$, $y = 0$, $x = a$, $x = b$ $(0 \le a < b)$ rotated about the y-axis: The enclosing cylinder has radius b and height $f(b)$. Its volume is $V_{\text{cyl}} = \pi b^2 f(b)$. The cavity is the solid generated by rotating the region bounded by $x = g(y)$ (inverse of f), $y = 0$, $y = f(b)$, and the y-axis? No, the cavity corresponds to the region to the left of the curve $x = g(y)$ from $y = 0$ to $y = f(a)$ (a cylinder of radius a) plus the region between $x = g(y)$ and $x = a$ from $y = f(a)$ to $y = f(b)$. Equivalently, the required volume is:

$$V = \pi b^2 f(b) - \pi a^2 f(a) - \pi \int_{f(a)}^{f(b)} [g(y)]^2\,dy.$$

If $a = 0$, this simplifies to $V = \pi b^2 f(b) - \pi \int_0^{f(b)} [g(y)]^2\,dy$.

EXEMPLE **Complementary Volume Technique**

The region bounded by $y = x^2$, $y = 0$, and $x = 2$ is rotated about the y-axis. Find the volume.

Method 1 (Washer method, $a = 0$): $x = \sqrt{y}$, limits $y = 0$ to $y = 4$. Outer radius $R = 2$, inner radius $r = \sqrt{y}$.

$$V = \pi \int_0^4 \left(2^2 - (\sqrt{y})^2\right) dy = \pi \int_0^4 (4 - y)\,dy = \pi\left[4y - \frac{y^2}{2}\right]_0^4 = \pi(16 - 8) = 8\pi.$$

Method 2 (Complementary volume): Enclosing cylinder: radius 2, height $f(2) = 4$. $V_{\text{cyl}} = \pi \cdot 2^2 \cdot 4 = 16\pi$. Cavity volume (solid generated by rotating region under $x = \sqrt{y}$ from $y = 0$ to 4 about y-axis): $V_{\text{cav}} = \pi \int_0^4 (\sqrt{y})^2\,dy = \pi \int_0^4 y\,dy = 8\pi$. Required volume $V = V_{\text{cyl}} - V_{\text{cav}} = 16\pi - 8\pi = 8\pi$.

WARNING **Common Pitfalls**

- Missing π: Always include π in the formula.
- Squaring the function incorrectly: The integrand is $[f(x)]^2$, not $f(x^2)$. For example, if $f(x) = \sin x$, then $[f(x)]^2 = \sin^2 x$, not $\sin(x^2)$.
- Wrong variable for y-axis rotation: If the curve is $y = f(x)$ and you rotate about the y-axis, do not write $\pi \int [f(x)]^2 \, dx$. You must express the radius in terms of y (washer method) or use the complementary volume method.
- Order of subtraction in washer method: Always subtract the square of the inner radius from the square of the outer radius: $R^2 - r^2$. Reversing them gives a negative volume.
- Limits of integration: For rotation about the y-axis, limits must be y-values. Find them by substituting x-bounds into $y = f(x)$.

4.6 Applications and Modelling Problems

GCE questions often present a real-world object (a bowl, a funnel, a reservoir) whose profile is given by a curve. You must set up the correct integral and evaluate it exactly.

WORKED EXAMPLE 12 **Modelling: A Parabolic Bowl**

A bowl is shaped by rotating the curve $y = \frac{1}{4}x^2$ for $0 \le x \le 4$ about the y-axis. Find the volume of the bowl.

SOLUTION

Solution: The curve is $x = 2\sqrt{y}$ (since $y = \frac{1}{4}x^2 \Rightarrow x^2 = 4y \Rightarrow x = 2\sqrt{y}$). The y-limits: when $x = 0$, $y = 0$; when $x = 4$, $y = 4$. Radius $R = 2\sqrt{y}$.

$$V = \pi \int_0^4 (2\sqrt{y})^2 \, dy = \pi \int_0^4 4y \, dy = 4\pi \left[\frac{y^2}{2} \right]_0^4 = 4\pi \cdot 8 = 32\pi \text{ units}^3.$$

WORKED EXAMPLE 13 **Modelling: A Funnel**

A funnel is formed by rotating the curve $y = \frac{1}{x}$ for $1 \le x \le 4$ about the x-axis. Find the volume of the funnel.

SOLUTION

Solution: Disc method about x-axis: $R = \frac{1}{x}$.

$$V = \pi \int_1^4 \left(\frac{1}{x}\right)^2 \, dx = \pi \int_1^4 x^{-2} \, dx = \pi \left[-\frac{1}{x}\right]_1^4 = \pi\left(-\frac{1}{4} + 1\right) = \frac{3\pi}{4} \text{ units}^3.$$

DID YOU KNOW? **Gabriel's Horn — A Paradox**

The solid obtained by rotating $y = 1/x$ for $x \geq 1$ about the x-axis has finite volume but infinite surface area. This means you could fill it with a finite amount of paint (π cubic units), but you could never paint its inner surface! It is a classic example of a counter-intuitive result in calculus.

4.7 Examination-Style Practice

EXERCISE 4 **GCE Type Questions**

1. The region bounded by $y = \sin x$, the x-axis, and $x = 0$, $x = \pi$ is rotated about the x-axis. Find the exact volume of the solid generated.
2. The curve $y^2 = 4x$ and the line $x = 4$ enclose a region. This region is rotated about the x-axis. Find the volume.
3. A reservoir has a cross-section given by $y = 2\sqrt{x}$ for $0 \leq x \leq 9$ (units in metres). The reservoir is formed by rotating this cross-section about the y-axis. Calculate its capacity in cubic metres, giving your answer as a multiple of π.
4. The region between $y = x^2$ and $y = 2x$ is rotated about the x-axis. Find the volume of the resulting solid.
5. A solid is generated by rotating the region bounded by $y = e^x$, $y = 0$, $x = 0$, and $x = \ln 2$ about the y-axis. Use the complementary volume method to find the exact volume.

SOLUTION **Exercise 1**

$V = \pi \int_0^{\pi} \sin^2 x \, dx = \pi \int_0^{\pi} \frac{1-\cos 2x}{2} \, dx = \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\pi} = \frac{\pi^2}{2}$.

SOLUTION **Exercise 2**

The curve is $y = \pm 2\sqrt{x}$. The region is symmetric about the x-axis. Using the upper half $y = 2\sqrt{x}$ from $x = 0$ to $x = 4$ and rotating about the x-axis gives a solid with volume $V = \pi \int_0^4 (2\sqrt{x})^2 \, dx = \pi \int_0^4 4x \, dx = 4\pi \left[\frac{x^2}{2}\right]_0^4 = 32\pi$.

SOLUTION **Exercise 3**

Curve: $x = \frac{y^2}{4}$ (since $y = 2\sqrt{x} \Rightarrow y^2 = 4x$). Limits: $y = 0$ to $y = 6$ (when $x = 9$, $y = 6$). Radius $R = \frac{y^2}{4}$. $V = \pi \int_0^6 \left(\frac{y^2}{4}\right)^2 dy = \pi \int_0^6 \frac{y^4}{16} dy = \frac{\pi}{16}\left[\frac{y^5}{5}\right]_0^6 = \frac{\pi}{16} \cdot \frac{7776}{5} = \frac{486\pi}{5}\ \text{m}^3$.

SOLUTION **Exercise 4**

Intersections: $x^2 = 2x \Rightarrow x = 0, 2$. On $[0, 2]$, $2x \geq x^2 \geq 0$. Outer radius $R = 2x$, inner radius $r = x^2$. $V = \pi \int_0^2 \left((2x)^2 - (x^2)^2\right) dx = \pi \int_0^2 (4x^2 - x^4) \, dx = \pi \left[\frac{4x^3}{3} - \frac{x^5}{5}\right]_0^2 = \pi\left(\frac{32}{3} - \frac{32}{5}\right) = \frac{64\pi}{15}$.

SOLUTION **Exercise 5**

Region: $y = e^x$ from $x = 0$ to $x = \ln 2$, $y = 0$. Rotate about y-axis. Cylinder enclosing it: radius $\ln 2$, height $e^{\ln 2} = 2$. Cylinder volume $V_{\text{cyl}} = \pi(\ln 2)^2 \cdot 2 = 2\pi(\ln 2)^2$. Volume of solid generated by region to the left of curve (i.e. $x = \ln y$ from $y = 1$ to $y = 2$): $V_{\text{left}} = \pi \int_1^2 (\ln y)^2 \, dy$. Integrate by parts twice: Let $u = (\ln y)^2$, $dv = dy \Rightarrow du = \frac{2 \ln y}{y} dy$, $v = y$. $\int (\ln y)^2 \, dy = y(\ln y)^2 - 2 \int \ln y \, dy = y(\ln y)^2 - 2(y \ln y - y) = y(\ln y)^2 - 2y \ln y + 2y$. Evaluate from 1 to 2: $2(\ln 2)^2 - 4 \ln 2 + 4 - (0 - 0 + 2) = 2(\ln 2)^2 - 4 \ln 2 + 2$. Thus $V_{\text{left}} = \pi\left(2(\ln 2)^2 - 4 \ln 2 + 2\right)$. Required volume $V = V_{\text{cyl}} - V_{\text{left}} = 2\pi(\ln 2)^2 - \pi\left(2(\ln 2)^2 - 4 \ln 2 + 2\right) = \pi(4 \ln 2 - 2) = 2\pi(2 \ln 2 - 1)$.

KEY POINTS **Summary of Volume Formulas**

- Disc about x-axis: $V = \pi \int_a^b [f(x)]^2 \, dx$
- Disc about y-axis: $V = \pi \int_c^d [g(y)]^2 \, dy$
- Washer about x-axis: $V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2\right) dx$
- Washer about y-axis: $V = \pi \int_c^d \left([G(y)]^2 - [H(y)]^2\right) dy$

- Complementary volume (about y-axis, $a = 0$): $V = \pi b^2 f(b) - \pi \int_0^{f(b)} [g(y)]^2 \, dy$
- Always: sketch, identify radii, choose correct variable, include π, give exact answers.

EXPERIMENT **Guided Demonstration: Deriving the Sphere Volume**

Take a semicircle of radius r: $y = \sqrt{r^2 - x^2}$ on $[-r, r]$. Rotate about the x-axis.

1. Write the Riemann sum for n discs of equal width $\Delta x = \frac{2r}{n}$.
2. Show that the sum is $\pi \sum_{i=1}^{n} (r^2 - x_i^2)\Delta x$.
3. Take the limit as $n \to \infty$ to obtain $\pi \int_{-r}^{r} (r^2 - x^2) \, dx$.
4. Evaluate the integral to confirm $V = \frac{4}{3}\pi r^3$.

This derivation is a classic GCE proof question.

5 SECTION 5: CENTROIDS AND CENTRES OF MASS BY INTEGRATION

In Section 4 we used integration to add up infinitely many thin discs and so obtain the volume of a solid of revolution. The very same idea — slicing a continuous object into thin pieces, computing the contribution of each piece, and adding the contributions with an integral — allows us to answer a new and very practical question: **where is the balancing point of a shape?** If you cut out a cardboard region bounded by a curve, on which single point could you balance it on the tip of a pencil? That point is the **centroid**, and integration finds it exactly. This section is also a direct bridge to your Mechanics papers, where centres of mass appear constantly.

5.1 From Discrete Masses to Continuous Distributions

Observation. Suppose three small masses are placed on a light rod along the x-axis: m_1 at x_1, m_2 at x_2, m_3 at x_3. From Mechanics, the rod balances at the point

$$\bar{x} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} = \frac{\sum m_i x_i}{\sum m_i}.$$

The numerator is the total moment of the masses about the origin; the denominator is the total mass. The centre of mass is simply "total moment divided by total mass".

Now imagine not three masses but a continuous sheet of metal (a **lamina**) occupying a region of the plane. Slice it into thin vertical strips. A strip at position x, of width dx, has a small mass dm and its moment about the origin is approximately $x \, dm$. Passing from the finite sum to the limit of infinitely many infinitely thin strips — exactly the Riemann-sum idea of Lesson 52 — replaces

$\sum$ by $\int$:

$$\bar{x} = \frac{\int x\,dm}{\int dm}.$$

If the lamina has constant density ρ (mass per unit area), then $dm = \rho\,dA$ where dA is the area of the strip. The constant ρ cancels in the fraction, so the balancing point depends only on the geometry. This geometric centre is called the **centroid**.

DEFINITION **Centroid and centre of mass**

The **centroid** of a plane region is its geometric centre: the point $(\bar{x}, \bar{y})$ where the region would balance if made from material of uniform (constant) density. For a lamina of constant density, the **centre of mass** and the centroid coincide, because the density factor cancels in "total moment ÷ total mass". We therefore use the two words interchangeably for uniform bodies. If the density varies, the centre of mass is given by $\bar{x} = \frac{\int x\rho(x,y)\,dA}{\int \rho(x,y)\,dA}$ (and similarly for $\bar{y}$), but the GCE syllabus focuses on the uniform case.

5.2 Centroid of the Region Under a Curve $y = f(x)$

Consider the region R bounded by the curve $y = f(x)$ (with $f(x) \geq 0$), the x-axis, and the lines $x = a$ and $x = b$. Take a thin vertical strip at position x, of width dx and height $f(x)$.

- **Area of the strip:** $dA = f(x)\,dx$, so the total area is $A = \int_a^b f(x)\,dx$.
- **Moment about the y-axis:** the whole strip sits at distance x from the y-axis, so its moment is $x\,f(x)\,dx$. Summing: $\int_a^b xf(x)\,dx$.
- **Moment about the x-axis:** the strip is a thin rectangle whose own centre is at height $\frac{1}{2}f(x)$, so its moment about the x-axis is $\frac{1}{2}f(x) \times f(x)\,dx = \frac{1}{2}[f(x)]^2\,dx$. Summing: $\int_a^b \frac{1}{2}[f(x)]^2\,dx$.

PROPERTY **Centroid formulas for the region under $y = f(x)$**

For the region under $y = f(x) \geq 0$ on $[a, b]$, with area $A = \int_a^b f(x)\,dx$:

$$\bar{x} = \frac{1}{A}\int_a^b x\,f(x)\,dx, \qquad \bar{y} = \frac{1}{A}\int_a^b \frac{1}{2}\,[f(x)]^2\,dx.$$

These formulas are examinable and should be quoted, then applied.

WARNING **The most common GCE error in this topic**

In the formula for $\bar{y}$, the integrand is $\frac{1}{2}[f(x)]^2$ — not $[f(x)]^2$ and not $\frac{1}{2}f(x)$. The $\frac{1}{2}$ comes from the fact that the centre of a vertical strip of height $f(x)$ is at height $\frac{1}{2}f(x)$. Forgetting it doubles your $\bar{y}$ and costs several marks. Write the factor $\frac{1}{2}$ down before you start integrating.

METHOD **Finding a centroid: the three-step routine**

1. Compute the area first: $A = \int_a^b f(x)\,dx$.
2. Compute the two moments: $M_y = \int_a^b xf(x)\,dx$ and $M_x = \int_a^b \frac{1}{2}[f(x)]^2\,dx$.
3. Divide: $\bar{x} = M_y/A$, $\bar{y} = M_x/A$. State the answer as a point $(\bar{x}, \bar{y})$.

Computing A first is essential: it is the common denominator of both coordinates, and examiners award a mark for it separately.

WORKED EXAMPLE 14 **Centroid of the triangle under a line**

Find the centroid of the region bounded by $y = x$, the x-axis and the line $x = 3$.

SOLUTION

Step 1 — Area.

$$A = \int_0^3 x\,dx = \left[\frac{x^2}{2}\right]_0^3 = \frac{9}{2}.$$

(Check: the region is a triangle of base 3 and height 3, area $\frac{1}{2}\cdot 3\cdot 3 = \frac{9}{2}$. ✓)

Step 2 — Moments.

$$M_y = \int_0^3 x\cdot x\,dx = \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = 9,$$

$$M_x = \int_0^3 \frac{1}{2}(x)^2\,dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^3 = \frac{9}{2}.$$

Step 3 — Divide.

$$\bar{x} = \frac{9}{9/2} = 2, \qquad \bar{y} = \frac{9/2}{9/2} = 1.$$

The centroid is $(2,\ 1)$. This agrees with the known result that the centroid of a triangle lies one third of the way up from the base: here $\bar{x} = 2$ (two thirds of the base from the vertex at the origin) and $\bar{y} = 1$ (one third of the height). ✓

WORKED EXAMPLE 15 **Centroid of the region under a parabola**

Find the centroid of the region bounded by $y = x^2$, the x-axis and the line $x = 2$.

SOLUTION

Step 1 — Area.

$$A = \int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

Step 2 — Moments.

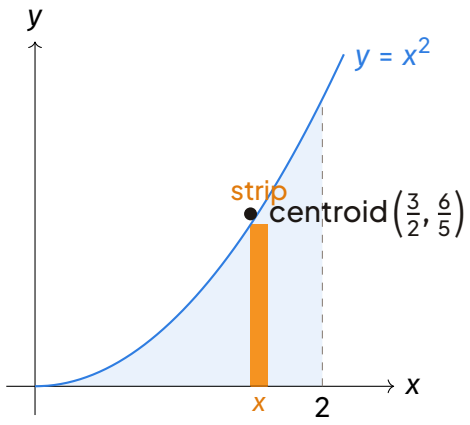
$$M_y = \int_0^2 x \cdot x^2\,dx = \int_0^2 x^3\,dx = \left[\frac{x^4}{4}\right]_0^2 = 4,$$

$$M_x = \int_0^2 \frac{1}{2}(x^2)^2\,dx = \frac{1}{2}\int_0^2 x^4\,dx = \frac{1}{2}\left[\frac{x^5}{5}\right]_0^2 = \frac{1}{2}\cdot\frac{32}{5} = \frac{16}{5}.$$

Step 3 — Divide.

$$\bar{x} = \frac{4}{8/3} = \frac{3}{2}, \qquad \bar{y} = \frac{16/5}{8/3} = \frac{16}{5}\times\frac{3}{8} = \frac{6}{5}.$$

The centroid is $\left(\frac{3}{2}, \frac{6}{5}\right)$. Note that $\bar{x} = \frac{3}{2} > \frac{2}{2} = 1$: the region has more area near $x = 2$ (where the parabola is tall), so the balance point shifts to the right of the midpoint — always check that your answer is plausible!



Region under $y = x^2$ on $[0, 2]$. The shaded vertical strip at position x has moment $x\,f(x)\,dx$ about the y-axis; the centroid (black dot) is the balance point of the whole region.

WORKED EXAMPLE 16 **Centroid under one arch of the sine curve**

Find the centroid of the region bounded by $y = \sin x$ and the x-axis for $0 \le x \le \pi$.

SOLUTION

Step 1 — Area.

$$A = \int_0^{\pi} \sin x \, dx = \left[-\cos x \right]_0^{\pi} = (-\cos \pi) - (-\cos 0) = 1 + 1 = 2.$$

Step 2 — Moments. For M_y, integrate by parts:

$$M_y = \int_0^{\pi} x \sin x \, dx = \left[-x \cos x \right]_0^{\pi} + \int_0^{\pi} \cos x \, dx = \pi + \left[\sin x \right]_0^{\pi} = \pi.$$

For M_x, use $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$:

$$M_x = \int_0^{\pi} \frac{1}{2} \sin^2 x \, dx = \frac{1}{4} \int_0^{\pi} (1 - \cos 2x) \, dx = \frac{1}{4} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} = \frac{\pi}{4}.$$

Step 3 — Divide.

$$\bar{x} = \frac{\pi}{2}, \qquad \bar{y} = \frac{\pi/4}{2} = \frac{\pi}{8}.$$

The centroid is $\left(\frac{\pi}{2}, \frac{\pi}{8} \right)$. Notice that $\bar{x} = \frac{\pi}{2}$ could have been predicted without any calculation: the arch is symmetric about $x = \frac{\pi}{2}$. We return to this idea below.

5.3 Region Between Two Curves

If the region is bounded above by $y = f(x)$ and below by $y = g(x)$ on $[a, b]$ (with $f(x) \geq g(x)$), each vertical strip runs from $g(x)$ to $f(x)$. Its height is $f(x) - g(x)$, so its area is $[f(x) - g(x)] \, dx$. The centre of the strip is at height $\frac{1}{2}[f(x) + g(x)]$. The moment about the x-axis is therefore

$$\frac{f(x) + g(x)}{2} [f(x) - g(x)] \, dx = \frac{1}{2} \left([f(x)]^2 - [g(x)]^2 \right) dx.$$

PROPERTY **Centroid of the region between two curves**

For the region between $y = f(x)$ (above) and $y = g(x)$ (below) on $[a, b]$, with $A = \int_a^b [f(x) - g(x)] \, dx$:

$$\bar{x} = \frac{1}{A} \int_a^b x \, [f(x) - g(x)] \, dx, \qquad \bar{y} = \frac{1}{A} \int_a^b \frac{1}{2} \left([f(x)]^2 - [g(x)]^2 \right) dx.$$

Setting $g(x) = 0$ recovers the single-curve formulas.

EXEMPLE **Quick check with two lines**

Region between $f(x) = 4$ and $g(x) = x$ on $[0, 4]$: $A = \int_0^4 (4 - x)\,dx = 8$; $M_y = \int_0^4 x(4 - x)\,dx = \left[2x^2 - \frac{x^3}{3}\right]_0^4 = \frac{32}{3}$; $M_x = \frac{1}{2}\int_0^4 (16 - x^2)\,dx = \frac{1}{2}\left[16x - \frac{x^3}{3}\right]_0^4 = \frac{64}{3}$. Hence $\bar{x} = \frac{4}{3}$, $\bar{y} = \frac{8}{3}$ — the centroid of a triangle with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, exactly the average of the vertices. ✓

5.4 Centre of Mass of a Uniform Solid of Revolution

Rotate the region under $y = f(x)$ on $[a, b]$ about the x-axis. The solid obtained (a vase, a cone, a bowl…) is symmetric about the x-axis, so its centre of mass lies on the axis: $\bar{y} = 0$ automatically. To find $\bar{x}$, slice the solid into thin discs perpendicular to the x-axis. The disc at position x has radius $f(x)$, thickness dx, hence volume $\pi[f(x)]^2\,dx$; for uniform density ρ, mass is $dm = \rho\pi[f(x)]^2\,dx$, and the disc's moment about the y-axis (through the origin) is $x \cdot \rho\pi[f(x)]^2\,dx$. The factors ρ and π cancel in the division.

PROPERTY **Centre of mass of a solid of revolution about the x-axis**

For the uniform solid generated by rotating $y = f(x)$ on $[a, b]$ about the x-axis:

$$\bar{x} = \frac{\displaystyle\int_a^b x\,[f(x)]^2\,dx}{\displaystyle\int_a^b [f(x)]^2\,dx}, \qquad \bar{y} = 0 \text{ (by symmetry).}$$

Note the denominator is the volume divided by π: $V = \pi\displaystyle\int_a^b [f(x)]^2 dx$, so equivalently $\bar{x} = \frac{\pi}{V}\displaystyle\int_a^b x[f(x)]^2 dx$.

WORKED EXAMPLE 17 **Centre of mass of a uniform cone**

A right circular cone of height h and base radius r is formed by rotating the line $y = \frac{r}{h}x$ on $[0, h]$ about the x-axis. Find its centre of mass.

SOLUTION

Denominator (volume integral).

$$\int_0^h \left(\frac{r}{h}x\right)^2 dx = \frac{r^2}{h^2}\left[\frac{x^3}{3}\right]_0^h = \frac{r^2 h}{3}.$$

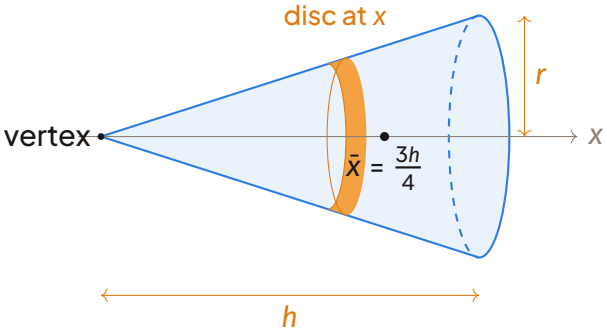
Numerator (moment integral).

$$\int_0^h x\left(\frac{r}{h}x\right)^2 dx = \frac{r^2}{h^2}\left[\frac{x^4}{4}\right]_0^h = \frac{r^2 h^2}{4}.$$

Divide.

$$\bar{x} = \frac{r^2h^2/4}{r^2h/3} = \frac{3h}{4}, \qquad \bar{y} = 0.$$

The centre of mass of a uniform solid cone is at $\frac{3h}{4}$ from the vertex (equivalently $\frac{h}{4}$ from the base) — a standard result worth memorising for Mechanics.



Uniform solid cone formed by rotating $y = \frac{r}{h}x$ about the x-axis. A disc slice at position x has volume $\pi y^2\,dx$ and moment $x\pi y^2\,dx$; summing and dividing gives $\bar{x} = \frac{3h}{4}$ from the vertex.

5.5 Exploiting Symmetry — a Time-Saving Exam Technique

KEY POINTS **Symmetry rule**

If a region or solid has an axis of symmetry, the centroid lies on that axis. If it has two axes of symmetry, the centroid is at their intersection and no integration is needed for either coordinate.

EXEMPLE **Symmetry in action**

- Rectangle, circle, ellipse: centroid at the geometric centre — immediately.
- Region under $y = \sin x$ on $[0, \pi]$: symmetric about $x = \frac{\pi}{2}$, so $\bar{x} = \frac{\pi}{2}$ with no calculation (only $\bar{y}$ needs integration).
- Any solid of revolution about the x-axis: $\bar{y} = 0$ by symmetry; only $\bar{x}$ is computed.
- Semicircular lamina of radius r (symmetric about the y-axis): $\bar{x} = 0$, and integration gives the standard result $\bar{y} = \frac{4r}{3\pi}$.

In the exam, state the symmetry argument explicitly ("by symmetry, $\bar{x} = \frac{\pi}{2}$") — it earns the

mark and halves your work.

5.6 Composite Laminas (Useful for Mechanics Papers)

Many Mechanics questions involve laminas built from standard shapes (rectangles, triangles, circles, semicircles). Since moments add, you may treat each part separately and combine:

$$\bar{x} = \frac{\sum A_i \bar{x}_i}{\sum A_i}, \qquad \bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i},$$

where a removed part (a hole) is counted with **negative** area. Standard centroids to know: rectangle — centre; triangle — one third of the height from the base; semicircle of radius r — $\frac{4r}{3\pi}$ from the diameter; solid cone — $\frac{h}{4}$ from the base.

EXEMPLE **Rectangle with a triangular corner removed**

A lamina is the rectangle $[0,6] \times [0,4]$ with the triangle of vertices $(4,0),(6,0),(6,4)$ removed. Rectangle: $A_1 = 24$, centre $(3,2)$. Triangle: $A_2 = \frac{1}{2}(2)(4) = 4$, centroid at the average of its vertices $\left(\frac{16}{3}, \frac{4}{3}\right)$, counted negatively:

$$\bar{x} = \frac{24(3) - 4\left(\frac{16}{3}\right)}{24 - 4} = \frac{72 - \frac{64}{3}}{20} = \frac{38}{15}, \qquad \bar{y} = \frac{24(2) - 4\left(\frac{4}{3}\right)}{20} = \frac{48 - \frac{16}{3}}{20} = \frac{32}{15}.$$

Both coordinates lie inside the remaining shape and are shifted away from the removed corner, as expected.

WARNING **Traps to avoid**

- Forgetting the factor $\frac{1}{2}$ in $\bar{y} = \frac{1}{A}\int \frac{1}{2}[f(x)]^2 dx$.
- Squaring $f(x)$ wrongly: for $y = \sin x$, $[f(x)]^2 = \sin^2 x$ — use the identity $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ before integrating.
- Mixing up the two moments: $M_y = \int x f(x)\,dx$ gives $\bar{x}$ (moment about the y-axis gives the x-coordinate).
- For a region between curves, using $[f(x) - g(x)]^2$ in $\bar{y}$ instead of $\frac{1}{2}\left([f(x)]^2 - [g(x)]^2\right)$. These are not the same!
- Giving $\bar{x}$ and $\bar{y}$ but forgetting to check plausibility: the centroid must lie inside (or on the boundary of) a convex region.

EXERCISE 5 **Consolidation**

1. Find the centroid of the region bounded by $y = 4 - x^2$ and the x-axis. (Use symmetry for $\bar{x}$.)
2. Find the centroid of the region between $y = x^2$ and $y = x$ on $[0, 1]$.
3. The region under $y = \sqrt{x}$ on $[0, 4]$ is rotated about the x-axis. Find the centre of mass of the solid formed.
4. A uniform lamina consists of a square of side 2 with a semicircle of radius 1 attached to one side. Locate its centre of mass.

SOLUTION **Exercise 1**

1. $A = \int_{-2}^{2}(4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \frac{32}{3}$. By symmetry $\bar{x} = 0$. $M_x = \frac{1}{2}\int_{-2}^{2}(4 - x^2)^2 dx = \frac{1}{2}\int_{-2}^{2}(16 - 8x^2 + x^4)\,dx = \frac{1}{2}\left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_{-2}^{2} = \frac{256}{15}$. Hence $\bar{y} = \frac{256/15}{32/3} = \frac{8}{5}$. Centroid: $\left(0, \frac{8}{5}\right)$.

2. $A = \int_0^1 (x - x^2)\,dx = \frac{1}{6}$. $M_y = \int_0^1 x(x - x^2)\,dx = \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 = \frac{1}{12}$, so $\bar{x} = \frac{1/12}{1/6} = \frac{1}{2}$. $M_x = \frac{1}{2}\int_0^1 (x^2 - x^4)\,dx = \frac{1}{2}\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \frac{1}{15}$, so $\bar{y} = \frac{1/15}{1/6} = \frac{2}{5}$. Centroid: $\left(\frac{1}{2}, \frac{2}{5}\right)$.

3. Denominator: $\int_0^4 x\,dx = 8$. Numerator: $\int_0^4 x \cdot x\,dx = \frac{64}{3}$. Hence $\bar{x} = \frac{64/3}{8} = \frac{8}{3}$, $\bar{y} = 0$.

4. Put the square on $[0, 2] \times [0, 2]$ and the semicircle on the side $x = 2$, bulging right. Square: $A_1 = 4$, centre $(1, 1)$. Semicircle: $A_2 = \frac{\pi}{2}$, centre $\left(2 + \frac{4}{3\pi}, 1\right)$. By symmetry $\bar{y} = 1$;

$$\bar{x} = \frac{4(1) + \frac{\pi}{2}\left(2 + \frac{4}{3\pi}\right)}{4 + \frac{\pi}{2}} = \frac{4 + \pi + \frac{2}{3}}{4 + \frac{\pi}{2}} = \frac{\frac{14}{3} + \pi}{4 + \frac{\pi}{2}} \approx 1.31.$$

DID YOU KNOW? **Why engineers care**

Civil engineers designing the beams and bridges that cross the rivers of Cameroon must know exactly where the centroid of each cross-section lies: bending stresses are computed about an axis through the centroid. The same formulas you have just used also locate the centre of gravity of aircraft wings, ship hulls and satellite components — a beautiful example of a Riemann sum keeping real structures safe.

6 SECTION 6: TRAPEZOID RULE AND THE MEAN VALUE THEOREM FOR INTEGRALS

Having mastered the computation of centroids and centres of mass by integration, we now turn to two practical tools that every scientist and engineer must possess: **numerical integration** when an exact antiderivative is unavailable, and the **mean value of a function** which gives a single representative number for a whole interval. Both ideas are built on the same geometric intuition that has guided us since the definite integral was introduced as a limit of Riemann sums.

6.1 Why Numerical Integration?

In earlier sections we evaluated definite integrals by finding an antiderivative F of f and applying the Fundamental Theorem of Calculus: $\int_a^b f(x)\,dx = F(b) - F(a)$. This works beautifully when f has an elementary antiderivative. However, in real-world situations we often encounter:

- Functions defined only by experimental data (e.g., temperature readings every hour, cocoa prices at the market each day, velocity of a vehicle recorded by a sensor).
- Functions that do have an antiderivative but it cannot be expressed in terms of elementary functions (e.g., e^{-x^2}, $\frac{\sin x}{x}$, $\sqrt{1+x^3}$).

In such cases we must approximate the integral numerically. The simplest and most widely taught method at this level is the **trapezoid rule** (also called the trapezium rule). It replaces the area under the curve by the sum of areas of trapezia whose top edges are chords of the curve.

6.2 The Trapezoid Rule

Consider a continuous function f on $[a, b]$. Divide $[a, b]$ into n equal strips of width

$$h = \frac{b-a}{n}.$$

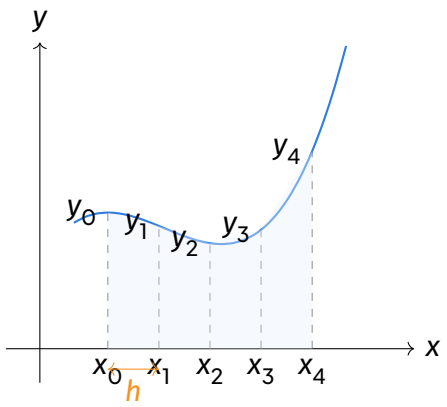
Let the ordinates be $y_0 = f(x_0), y_1 = f(x_1), \dots, y_n = f(x_n)$ where $x_k = a + kh$. The area of the k-th trapezium (between x_{k-1} and x_k) is $\frac{h}{2}(y_{k-1} + y_k)$. Summing from $k = 1$ to n gives the trapezoid rule approximation:

PROPERTY **Trapezoid Rule with n Equal Strips**

If f is continuous on $[a, b]$ and $h = \frac{b-a}{n}$, then

$$\int_a^b f(x)\,dx \approx \frac{h}{2}\left[\, y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n \,\right],$$

where $y_k = f(x_k)$ and $x_k = a + kh$ for $k = 0, 1, \dots, n$.



Trapezoid rule with $n = 4$ strips: the chords approximate the curve, and the shaded trapezia approximate the area under the curve.

6.3 Organising Computations: The Ordinate Table

To avoid errors, especially in examination conditions, always set up an **ordinate table**. This is a simple two-row table listing the x-values and the corresponding y-values.

METHOD **Ordinate Table for the Trapezoid Rule**

1. Determine a, b and the number of strips n (or the strip width h).
2. Compute $h = \frac{b-a}{n}$.
3. List the x-values: $x_0 = a,\ x_1 = a + h,\ \ldots,\ x_n = b$.
4. Evaluate $y_k = f(x_k)$ for each x_k (or read them from a given data table).
5. Write the ordinates in a row, then apply the formula:

$$\text{Area} \approx \frac{h}{2}\left[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\right].$$

WARNING **First and Last Ordinates Count Once!**

A very common mistake is to multiply all ordinates by 2. Remember: the first ordinate y_0 and the last ordinate y_n are multiplied by 1 (i.e., not doubled), while every interior ordinate $y_1, \ldots, y_{n-1}$ is multiplied by 2.

EXEMPLE **Trapezoid Rule from a Formula**

Use the trapezoid rule with $n = 4$ strips to approximate $\int_1^3 \frac{1}{x}\,dx$.

WORKED EXAMPLE 18 **Solution**

Here $a = 1$, $b = 3$, $n = 4$, so $h = \frac{3-1}{4} = 0.5$.

x	1	1.5	2	2.5	3
$y = 1/x$	1.0000	0.6667	0.5000	0.4000	0.3333

$$\int_1^3 \frac{1}{x}\,dx \approx \frac{0.5}{2}\left[1.0000 + 2(0.6667 + 0.5000 + 0.4000) + 0.3333\right]$$
$$= 0.25\left[1.0000 + 2(1.5667) + 0.3333\right]$$
$$= 0.25\left[1.0000 + 3.1334 + 0.3333\right]$$
$$= 0.25 \times 4.4667 = 1.1167.$$

The exact value is $\ln 3 \approx 1.0986$, so the approximation overestimates by about 0.0181.

EXEMPLE **Trapezoid Rule from Experimental Data**

The speed v (in m s^{-1}) of a car was recorded every 2 seconds:

t (s)	0	2	4	6	8	10	12
v (m s^{-1})	0	5.2	9.8	12.1	13.0	12.5	11.0

Estimate the distance travelled in the first 12 seconds using the trapezoid rule.

WORKED EXAMPLE 19 **Solution**

Here the data are given directly; no formula for $v(t)$ is needed. We have $a = 0$, $b = 12$, $n = 6$ strips, so $h = 2$.

$$\text{Distance} \approx \frac{2}{2}\left[0 + 2(5.2 + 9.8 + 12.1 + 13.0 + 12.5) + 11.0\right]$$
$$= 1 \times \left[0 + 2(52.6) + 11.0\right]$$
$$= 105.2 + 11.0 = 116.2 \text{ m}.$$

6.4 Accuracy and Error Behaviour

The trapezoid rule is exact for linear functions. For curved graphs, the error depends on the **concavity**:

- If f is **concave up** ($f''(x) > 0$) on $[a, b]$, the chords lie above the curve ⇒ trapezoid rule **overestimates** the integral.
- If f is **concave down** ($f''(x) < 0$) on $[a, b]$, the chords lie below the curve ⇒ trapezoid rule **underestimates** the integral.

Increasing n (decreasing h) reduces the error. The error is proportional to h^2 (or $1/n^2$), so doubling n roughly quarters the error.

WARNING **Exam Tip**

Always state the value of h explicitly. Keep at least 4 decimal places in intermediate ordinate values to avoid rounding errors that accumulate in the final sum.

6.5 The Mean Value of a Function

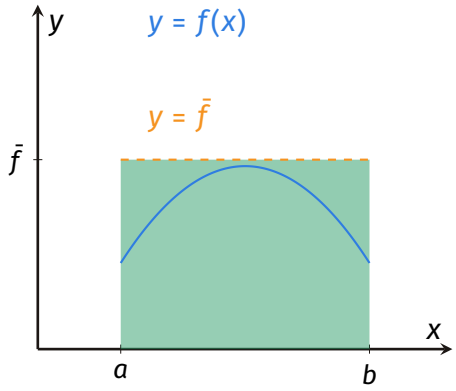
If a function f is continuous on $[a, b]$, its **mean value** (or average value) on that interval is the constant height $\bar{f}$ such that a rectangle of width $b - a$ and height $\bar{f}$ has the same area as the region under the curve.

DEFINITION **Mean Value of a Function**

The mean value of a continuous function f on the interval $[a, b]$ is

$$\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Geometrically, $\bar{f}$ is the height of the rectangle with base $[a, b]$ whose area equals $\int_a^b f(x)\,dx$.



The mean value $\bar{f}$ is the height of the rectangle (green) whose area equals the area under the curve (blue). The two shaded regions have equal area.

EXEMPLE **Mean Value of a Polynomial**

Find the mean value of $f(x) = x^2$ on $[0, 2]$.

WORKED EXAMPLE 20 **Solution**

$$\bar{f} = \frac{1}{2-0}\int_0^2 x^2\,dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2 = \frac{1}{2}\cdot\frac{8}{3} = \frac{4}{3}.$$

The rectangle of width 2 and height $4/3$ has area $8/3$, same as the integral.

6.6 The Mean Value Theorem for Integrals

The Mean Value Theorem for Integrals guarantees that the mean value is actually attained by the function at some point in the interval (provided f is continuous).

PROPERTY **Mean Value Theorem for Integrals**

If f is continuous on $[a, b]$, then there exists at least one number $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx = \bar{f}.$$

KEY POINTS **Key Points**

- The theorem is an existence theorem: it tells us such a c exists, but does not give a formula to find it.
- Graphically: the horizontal line $y = \bar{f}$ cuts the curve $y = f(x)$ at least once between a and b.
- If f is not continuous, the theorem may fail (e.g., a function with a jump discontinuity might never equal its mean value).

EXEMPLE **Applying the Mean Value Theorem**

Let $f(x) = x^2$ on $[0, 2]$. We found $\bar{f} = 4/3$. Find a value $c \in [0, 2]$ such that $f(c) = \bar{f}$.

WORKED EXAMPLE 21 **Solution**

We need $c^2 = \frac{4}{3}$, so $c = \frac{2}{\sqrt{3}} \approx 1.155$. Indeed $c \in [0, 2]$.

6.7 Applications and Combined Problems

The mean value and the trapezoid rule often appear together in GCE questions. Typical applications include:

- **Average speed:** $\bar{v} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v(t)\,dt$.
- **Average temperature:** $\bar{T} = \frac{1}{24} \int_0^{24} T(t)\,dt$ (if t is in hours).
- **Average price of cocoa:** If the price per kilogram varies over a season, the mean price gives a fair single figure for contracts.

EXEMPLE **Combined Problem: Trapezoid Estimate of Mean Value**

The temperature T (in $^\circ$C) in a greenhouse was recorded over a 12-hour period:

t (h)	0	2	4	6	8	10	12
T ($^\circ$C)	18	20	23	25	24	22	20

1. Use the trapezoid rule with 6 strips to estimate $\int_0^{12} T(t)\,dt$.
2. Hence estimate the mean temperature over the 12-hour period.

WORKED EXAMPLE 22 **Solution**

Step 1: $h = 2$, $n = 6$.

$$\int_0^{12} T(t)\,dt \approx \frac{2}{2}\left[18 + 2(20 + 23 + 25 + 24 + 22) + 20\right]$$
$$= 1 \times \left[18 + 2(114) + 20\right]$$
$$= 18 + 228 + 20 = 266 \text{ degree-hours.}$$

Step 2: Mean temperature $\bar{T} = \frac{1}{12-0} \times 266 = \frac{266}{12} \approx 22.17^{\circ}\text{C}$.

EXEMPLE **GCE-Style Problem: Existence of c**

The velocity of a particle is given by $v(t) = 3t^2 - 12t + 9$ for $0 \le t \le 4$.

1. Find the mean velocity on $[0, 4]$.
2. Show that there is a time $c \in [0, 4]$ when the instantaneous velocity equals the mean velocity, and find c.

WORKED EXAMPLE 23 **Solution**

1. Mean velocity $\bar{v} = \frac{1}{4}\int_0^4 (3t^2 - 12t + 9)\,dt$.

$$\int_0^4 (3t^2 - 12t + 9)\,dt = \left[t^3 - 6t^2 + 9t\right]_0^4 = (64 - 96 + 36) - 0 = 4.$$

So $\bar{v} = \frac{4}{4} = 1\,\text{m s}^{-1}$.

2. Since v is continuous (polynomial), the Mean Value Theorem for Integrals guarantees a $c \in [0, 4]$ with $v(c) = 1$. Solve $3c^2 - 12c + 9 = 1 \Rightarrow 3c^2 - 12c + 8 = 0$.

$$c = \frac{12 \pm \sqrt{144 - 96}}{6} = \frac{12 \pm \sqrt{48}}{6} = 2 \pm \frac{2\sqrt{3}}{3}.$$

Both roots lie in $[0, 4]$: $c_1 = 2 - \frac{2\sqrt{3}}{3} \approx 0.845$, $c_2 = 2 + \frac{2\sqrt{3}}{3} \approx 3.155$.

DID YOU KNOW? **Did You Know?**

The trapezoid rule is the simplest of the Newton–Cotes formulas. A more accurate method using parabolic arcs is Simpson's rule, which requires an even number of strips and gives exact results for polynomials up to degree 3. In Cameroon, numerical integration is used by

engineers designing the Lom Pangar dam spillways and by agronomists estimating cocoa yield from discrete field samples.

EXERCISE 6 Practice Problems

1. Use the trapezoid rule with $n = 5$ to approximate $\int_0^1 e^{-x^2}\,dx$. (Use $e^{-0.04} = 0.9608$, $e^{-0.16} = 0.8521$, $e^{-0.36} = 0.6977$, $e^{-0.64} = 0.5273$, $e^{-1} = 0.3679$.)

2. The table gives the rate of flow R (in m^3/s) of a river at 2-hour intervals:

t (h)	0	2	4	6	8	10
R (m^3/s)	12	15	18	20	17	14

   Estimate the total volume of water that passed in the 10-hour period. (Give your answer in m^3.)

3. Find the mean value of $f(x) = \sin x$ on $[0, \pi]$. Verify the Mean Value Theorem for Integrals by finding c.

4. A farmer records the price of cocoa (in FCFA/kg) at the start of each month for 6 months: 1200, 1250, 1300, 1280, 1320, 1350. Assuming the price varies linearly between recordings, estimate the average price over the 6-month period using the trapezoid rule. (Take the time interval as 0 to 5 months, with 5 strips of width 1 month.)

SOLUTION Exercise 1

$h = 0.2$. Ordinates: $y_0 = 1$, $y_1 = 0.9608$, $y_2 = 0.8521$, $y_3 = 0.6977$, $y_4 = 0.5273$, $y_5 = 0.3679$.

$$\int_0^1 e^{-x^2}\,dx \approx \frac{0.2}{2}[1+2(0.9608+0.8521+0.6977+0.5273)+0.3679] = 0.1[1+2(3.0379)+0.3679] = 0.1[7.4437] = 0$$

SOLUTION Exercise 2

$h = 2$ hours, $n = 5$ strips.

$$\int_0^{10} R(t)\,dt \approx \frac{2}{2}[12 + 2(15 + 18 + 20 + 17) + 14] = 1[12 + 2(70) + 14] = 166\ \text{m}^3/\text{s}\cdot\text{h}.$$

Since 1 h = 3600 s, the volume is $166 \times 3600 = 597\,600\ \text{m}^3$.

SOLUTION **Exercise 3**

$\bar{f} = \frac{1}{\pi}\int_0^{\pi} \sin x\,dx = \frac{1}{\pi}[-\cos x]_0^{\pi} = \frac{2}{\pi}$. Solve $\sin c = \frac{2}{\pi} \Rightarrow c = \arcsin(2/\pi) \approx 0.69$ or $c = \pi - \arcsin(2/\pi) \approx 2.45$. Both in $[0, \pi]$.

SOLUTION **Exercise 4**

Time interval: 0 to 5 months (start of month 1 to start of month 6), $n = 5$ strips, $h = 1$ month. Prices: $y_0 = 1200$, $y_1 = 1250$, $y_2 = 1300$, $y_3 = 1280$, $y_4 = 1320$, $y_5 = 1350$.

Integral $\approx \frac{1}{2}[1200+2(1250+1300+1280+1320)+1350] = 0.5[1200+2(5150)+1350] = 0.5[12850] = 6425$ FCFA

Mean price $= \frac{1}{5} \times 6425 = 1285$ FCFA/kg. (Note: The interval length is 5 months, not 6, because the 6 recordings are at the starts of months 1 through 6.)

6.8 Summary of Key Formulas

KEY POINTS **Formulas to Remember**

- Trapezoid rule: $\int_a^b f(x)\,dx \approx \frac{h}{2}\left[y_0 + 2\sum_{k=1}^{n-1} y_k + y_n\right]$, $h = \frac{b-a}{n}$.
- Mean value: $\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx$.
- Mean Value Theorem for Integrals: If f continuous on $[a, b]$, $\exists c \in [a, b]$ such that $f(c) = \bar{f}$.

With these tools you can tackle any GCE question on numerical integration or mean values. Practice setting up ordinate tables neatly, and always interpret your results in the context of the problem — whether it's the distance travelled by a taxi in Douala, the average temperature in Ngaoundéré, or the fair price of cocoa in the South-West region.

7 INTEGRATION ACTIVITY

7.1 Aims of the activity

This activity is a **modelling project** that brings together all the major skills of the chapter in one realistic engineering task. By the end of the activity, you should be able to:

- model a real container as a **solid of revolution** and compute its exact capacity using $V = \pi \int_a^b y^2\,dx$;
- estimate the same quantity numerically with the **trapezoid rule** and comment on the accuracy of the approximation;
- locate the **centre of mass** of a solid of revolution by integration, and interpret it physically as a balancing point;
- use the trapezoid rule on discrete data (a table of measurements) to estimate a total consumption, and deduce a **mean value**;
- communicate a complete mathematical argument: sketch, computation, units, conclusion and critical discussion.

WARNING **Working in groups**

Work in groups of three or four. Every member must be able to explain each step: at the GCE Advanced Level, you will be alone in front of the paper! Keep all units in metres, cubic metres and litres ($1\ \text{m}^3 = 1000$ litres).

7.2 Situation

The neighbourhood of Nkwen, on the slopes of Bamenda, suffers from irregular water supply during the dry season. A small local NGO plans to build a **horizontal reservoir with a curved (bowl-shaped) profile**, half-buried on the hillside, to store rainwater collected in the rainy season.

The engineer proposes the following design. In a vertical plane through the axis of the reservoir, the upper wall of the container is modelled by the curve

$$y = 2\sqrt{x}, \qquad 0 \le x \le 9,$$

where x is the distance in metres measured along the axis from the narrow (closed) end, and y is the radius of the circular cross-section at that point, also in metres. The reservoir is the **solid of revolution** obtained by rotating this curve about the x-axis between $x = 0$ and $x = 9$. The wide end at $x = 9$ is closed by a flat circular plate.

Because the reservoir is heavy when full, it must rest on a **support cradle** placed exactly under the **centre of mass** of the solid (assumed of uniform density), so that the structure does not tip over.

During the first week of a field study, a flow meter on the outlet pipe gave the following readings of the outflow rate $r(t)$, in cubic metres per day, where t is the time in days from Monday 6:00:

Day t	0	1	2	3	4	5	6
Outflow $r(t)$ (m^3/day)	12	15	18	14	10	9	11

The NGO wants to know: Is the reservoir large enough to supply the neighbourhood for one week of dry season, starting full? Your group is the consulting mathematics team.

7.3 Tasks

1. **Sketch the design.** Draw the curve $y = 2\sqrt{x}$ for $0 \le x \le 9$, shade the region to be rotated, and sketch the resulting solid of revolution. What is the radius of the circular plate closing the wide end?

2. **Exact capacity.** Show that the exact volume of the reservoir is $V = \pi \int_0^9 4x\,dx$, and compute it exactly. Convert the capacity to litres.

3. **Numerical check.** A worker can only measure the radius at seven equally spaced positions: $x = 0, 1.5, 3, 4.5, 6, 7.5, 9$. Compute the cross-sectional area $A(x) = \pi y^2$ at each position, then use the **trapezoid rule** with these seven ordinates to estimate $V = \int_0^9 A(x)\,dx$. Compare with the exact value: give the absolute and percentage error, and comment on the result in the light of the shape of $A(x)$.

4. **Position of the support cradle.** The centre of mass of a solid of revolution about the x-axis lies on the axis at
$$\bar{x} = \frac{\displaystyle\int_0^9 x\,\pi y^2\,dx}{\displaystyle\int_0^9 \pi y^2\,dx}.$$
Compute $\bar{x}$ exactly. How far from the narrow end must the cradle be placed? Is it before or after the midpoint of the reservoir? Justify physically.

5. **Weekly consumption.** Using the outflow table and the trapezoid rule with step $h = 1$ day, estimate the total volume of water consumed during the six recorded days, $Q = \int_0^6 r(t)\,dt$.

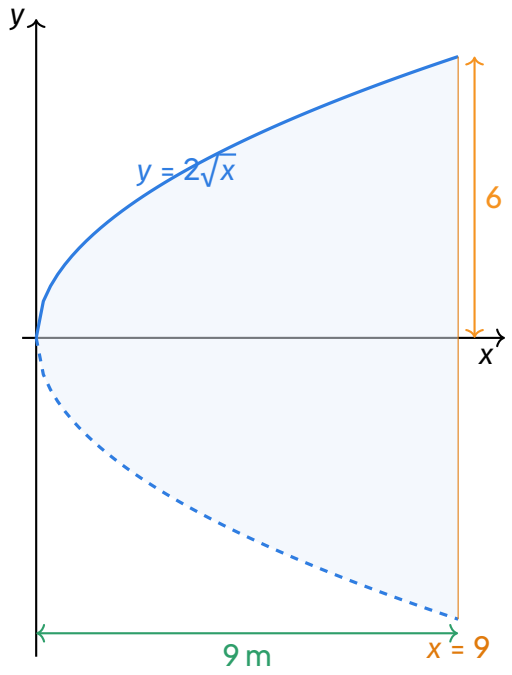
6. **Mean daily outflow.** Deduce an estimate of the **mean value** of the outflow rate over the week, $\bar{r} = \frac{1}{6}\int_0^6 r(t)\,dt$. State the Mean Value Theorem for integrals and explain what it guarantees here, assuming r is continuous.

7. **Verdict and discussion.** Starting full, is the reservoir sufficient for the week? Compute the margin (volume left or shortfall). Discuss two limitations of your model (for example: evaporation, leakage, the trapezoid approximation of r, or the idealised shape $y = 2\sqrt{x}$), and suggest one improvement to the design or the data collection.

8. **Presentation.** Prepare a five-minute group presentation with your sketches, the key integrals, and your recommendation to the NGO.

7.4 Model answer

1. Sketch. The curve $y = 2\sqrt{x}$ starts at the origin and rises to $y = 2\sqrt{9} = 6$ at $x = 9$; it is concave (it flattens as x grows). Rotating about the x-axis gives a bowl-shaped (paraboloid-like) reservoir, narrow at $x = 0$ and wide at $x = 9$. The closing plate at $x = 9$ is a disc of radius 6 m.



The reservoir as a solid of revolution of $y = 2\sqrt{x}$ about the x-axis.

2. Exact capacity. Since $y = 2\sqrt{x}$, we have $y^2 = 4x$, so the cross-section at position x is a disc of area $\pi y^2 = 4\pi x$. Hence

$$V = \pi \int_0^9 y^2 \, dx = \pi \int_0^9 4x \, dx = \pi \left[2x^2\right]_0^9 = \pi(162 - 0) = 162\pi \text{ m}^3.$$

Numerically $V \approx 508.94$ m^3, i.e. about 508 938 litres.

3. Trapezoid check. With $A(x) = \pi y^2 = 4\pi x$ and step $h = 1.5$:

x	0	1.5	3	4.5	6	7.5	9
$A(x) = 4\pi x$	0	6π	12π	18π	24π	30π	36π

The trapezoid rule with $n = 6$ strips gives

$$V \approx \frac{h}{2}\left[A_0 + A_6 + 2(A_1 + A_2 + A_3 + A_4 + A_5)\right].$$

Factoring out π from every ordinate:

$$V \approx \frac{1.5}{2}\,\pi\,[0 + 36 + 2(6 + 12 + 18 + 24 + 30)] = 0.75\,\pi\,[36 + 2 \times 90] = 0.75\,\pi \times 216 = 162\pi\ \text{m}^3.$$

The estimate is exactly 162π: the absolute error is 0 and the percentage error is 0%. This is not a coincidence: $A(x) = 4\pi x$ is a linear function, and the trapezoid rule is exact for linear functions, because on each strip the chord (the top of the trapezium) coincides with the graph itself.

WARNING **Where the concavity argument really applies**

The concavity remark concerns the radius $y = 2\sqrt{x}$, which is concave, not the area $A(x) = 4\pi x$, which is linear. Had we applied the trapezoid rule to the radius values and then squared the result, we would have obtained an underestimate of the true profile (the chords of a concave curve lie below the curve). Always apply the trapezoid rule to the quantity that actually appears under the integral sign — here $A(x) = \pi y^2$.

4. Support cradle. Compute the numerator:

$$\int_0^9 x\,\pi y^2\,dx = \pi \int_0^9 x \cdot 4x\,dx = \pi \int_0^9 4x^2\,dx = \pi\left[\frac{4x^3}{3}\right]_0^9 = \pi \cdot \frac{4 \times 729}{3} = 972\pi.$$

The denominator is the volume already found: $\int_0^9 \pi y^2\,dx = 162\pi$. Hence

$$\bar{x} = \frac{972\pi}{162\pi} = 6\ \text{m}.$$

The cradle must be placed 6 m from the narrow end, i.e. beyond the geometric midpoint (4.5 m). This is physically sensible: the reservoir is wider — and therefore heavier per unit length — towards the wide end, so the balancing point shifts towards $x = 9$.

5. Weekly consumption. With $h = 1$ and the seven readings 12, 15, 18, 14, 10, 9, 11:

$$Q \approx \frac{1}{2}\,[12 + 11 + 2(15 + 18 + 14 + 10 + 9)] = \frac{1}{2}\,[23 + 2 \times 66] = \frac{155}{2} = 77.5\ \text{m}^3.$$

6. Mean daily outflow.

$$\bar{r} = \frac{1}{6}\int_0^6 r(t)\,dt \approx \frac{77.5}{6} \approx 12.9\ \text{m}^3/\text{day}.$$

The **Mean Value Theorem for integrals** states: if r is continuous on $[a, b]$, then there exists at least one $c \in [a, b]$ such that

$$r(c) = \frac{1}{b - a}\int_a^b r(t)\,dt.$$

Here it guarantees that at some moment of the week the outflow rate was actually about $12.9\ \text{m}^3/\text{day}$ — the average is a value genuinely attained by the flow, not just an arithmetic artefact.

7. Verdict. Weekly consumption $\approx 77.5\ \text{m}^3$ against a capacity of $162\pi \approx 508.94\ \text{m}^3$. The week consumes only about $\frac{77.5}{508.94} \approx 15\%$ of the capacity, leaving a margin of roughly $508.94 - 77.5 \approx$

$431\,\text{m}^3$. **The reservoir is amply sufficient** — in fact, at this rate it could supply the neighbourhood for about $\dfrac{508.94}{77.5} \approx 6.6$ weeks.

Limitations. (i) The trapezoid estimate of Q uses only one reading per day; night-time peaks are invisible — more frequent readings would improve it. (ii) The model ignores evaporation and leakage, which reduce the effective capacity. (iii) The idealised profile $y = 2\sqrt{x}$ may differ from the structure actually built. Improvement: record the outflow every two hours and repeat the trapezoid rule with $h = \frac{1}{12}$ day, and coat the interior of the reservoir to reduce leakage.

8. Presentation. Each group presents: the sketch, the exact integral $V = 162\pi\,\text{m}^3$, the trapezoid verification, the cradle position $\bar{x} = 6\,\text{m}$, the consumption $Q \approx 77.5\,\text{m}^3$ with mean $\bar{r} \approx 12.9\,\text{m}^3/\text{day}$, and the final recommendation to the NGO.

KEY POINTS **What this activity consolidates**

- Volume of revolution about the x-axis: $V = \pi \displaystyle\int_a^b y^2\,dx$.
- Centre of mass on the axis: $\bar{x} = \dfrac{\displaystyle\int_a^b x\,y^2\,dx}{\displaystyle\int_a^b y^2\,dx}$.
- Trapezoid rule: $\displaystyle\int_a^b f(x)\,dx \approx \frac{h}{2}\left[y_0 + y_n + 2\sum_{i=1}^{n-1} y_i\right]$ — exact for linear f.
- Mean value of a function: $\bar{f} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$, and the Mean Value Theorem guarantees this value is attained when f is continuous.

8 METHODS AND WORKED EXAMPLES

9 THE DEFINITE INTEGRAL AS A LIMIT OF RIEMANN SUMS

9.1 Definition and Interpretation

DEFINITION **Partition and Riemann Sum**

Let f be a function defined on a closed interval $[a, b]$. A partition P of $[a, b]$ is a finite set of points $a = x_0 < x_1 < \cdots < x_n = b$. The norm of the partition is $\|P\| = \max_{1 \le i \le n}(x_i - x_{i-1})$. For each subinterval $[x_{i-1}, x_i]$, choose a sample point $x_i^* \in [x_{i-1}, x_i]$. The Riemann sum of f associated with P and the sample points is

$$S(P, f) = \sum_{i=1}^{n} f(x_i^*)\,(x_i - x_{i-1}).$$

If f is continuous on $[a, b]$, the limit of Riemann sums as $\|P\| \to 0$ exists and is independent of the choice of partitions and sample points.

PROPERTY **Definite Integral as a Limit of Riemann Sums**

If f is continuous on $[a, b]$, the definite integral of f from a to b is defined by

$$\int_a^b f(x)\,dx = \lim_{\|P\| \to 0} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i,$$

where $\Delta x_i = x_i - x_{i-1}$. In particular, for a regular partition with n equal subintervals of width $\Delta x = \frac{b-a}{n}$ and right endpoints $x_k = a + k\Delta x$,

$$\int_a^b f(x)\,dx = \lim_{n \to \infty} \sum_{k=1}^{n} f(a + k\Delta x)\,\Delta x.$$

9.2 Method: Evaluating a Definite Integral from the Definition

METHOD **From Riemann sum to definite integral**

To compute $\int_a^b f(x)\,dx$ directly from the definition using a regular partition and right endpoints:

1. Divide $[a, b]$ into n equal sub-intervals of width $\Delta x = \dfrac{b-a}{n}$.
2. Choose the right endpoints $x_k = a + k\Delta x$, $k = 1, \ldots, n$.
3. Form the Riemann sum $S_n = \sum_{k=1}^{n} f(x_k)\,\Delta x$.
4. Simplify S_n using the standard sums

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2.$$

5. Take the limit: $\int_a^b f(x)\,dx = \lim_{n\to\infty} S_n$.

Reflex: conversely, a limit of the form $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right)$ equals $\int_0^1 f(x)\,dx$. More generally, $\lim_{n\to\infty} \frac{b-a}{n}\sum_{k=1}^{n} f\left(a + \frac{k(b-a)}{n}\right) = \int_a^b f(x)\,dx$.

WORKED EXAMPLE 24 **Riemann sum for $f(x) = x^2$ on $[0, 1]$**

Using Riemann sums with n equal sub-intervals and right endpoints, show that $\int_0^1 x^2\,dx = \frac{1}{3}$.

SOLUTION

Here $a = 0$, $b = 1$, so $\Delta x = \dfrac{1}{n}$ and the right endpoints are $x_k = \dfrac{k}{n}$. The Riemann sum is

$$S_n = \sum_{k=1}^{n} f(x_k)\,\Delta x = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \cdot \frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n} k^2.$$

Using $\sum_{k=1}^{n} k^2 = \dfrac{n(n+1)(2n+1)}{6}$:

$$S_n = \frac{n(n+1)(2n+1)}{6n^3} = \frac{(n+1)(2n+1)}{6n^2}.$$

Expanding the numerator: $(n+1)(2n+1) = 2n^2 + 3n + 1$, so

$$S_n = \frac{2n^2+3n+1}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

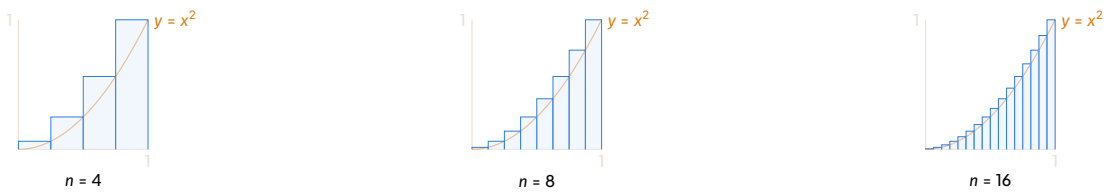
Taking the limit as $n \to \infty$, the terms $\frac{1}{2n}$ and $\frac{1}{6n^2}$ tend to 0, hence

$$\int_0^1 x^2\,dx = \lim_{n\to\infty} S_n = \frac{1}{3}.$$

Check: by antidifferentiation, $\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$. ✓

EXPERIMENT **Visualising Riemann Sums**

The figure below shows the right-endpoint Riemann sums for $f(x) = x^2$ on $[0,1]$ with $n = 4, 8, 16$. As n increases, the rectangles fill the region under the curve, illustrating the convergence of the sum to the exact area $\frac{1}{3}$.



Right-endpoint Riemann sums for $y = x^2$ on $[0,1]$ with increasing n.

10 AREAS BOUNDED BY THE X-AXIS AND BETWEEN CURVES

10.1 Area between a Curve and the x-axis

METHOD **Area bounded by the x-axis**

To find the area enclosed by $y = f(x)$, the x-axis, and the lines $x = a$, $x = b$:

1. Sketch the curve and locate its intersections with the x-axis on $[a, b]$ (solve $f(x) = 0$).
2. Determine the sign of f on each sub-interval.
3. On intervals where $f(x) \geq 0$: area $= \int f(x)\,dx$. Where $f(x) \leq 0$: area $= -\int f(x)\,dx$ (or $\int |f(x)|\,dx$).
4. Add the positive contributions of each piece.

WARNING **The sign trap**

$\int_a^b f(x)\,dx$ gives the algebraic (signed) area. If the curve crosses the axis, a single integral may give 0 or a small number while the true area is large. Always split at the zeros of f.

WORKED EXAMPLE 25 **Area under $y = x^2 - 4$ between $x = 0$ and $x = 3$**

Find the area of the region bounded by the curve $y = x^2 - 4$, the x-axis and the lines $x = 0$ and $x = 3$.

SOLUTION

Step 1: zeros of f. $x^2 - 4 = 0 \iff x = \pm 2$. On $[0, 3]$, the curve crosses the axis at $x = 2$.
Step 2: sign of f. For $0 \le x < 2$, $x^2 - 4 < 0$ (curve below the axis). For $2 < x \le 3$, $x^2 - 4 > 0$ (curve above).
Step 3: split the integral.

$$A = -\int_0^2 (x^2 - 4)\,dx + \int_2^3 (x^2 - 4)\,dx.$$

An antiderivative is $F(x) = \frac{x^3}{3} - 4x$.
First piece:

$$\int_0^2 (x^2 - 4)\,dx = \left[\frac{x^3}{3} - 4x\right]_0^2 = \frac{8}{3} - 8 = -\frac{16}{3}.$$

Second piece:

$$\int_2^3 (x^2 - 4)\,dx = \left(\frac{27}{3} - 12\right) - \left(\frac{8}{3} - 8\right) = -3 + \frac{16}{3} = \frac{7}{3}.$$

Step 4: total area.

$$A = \frac{16}{3} + \frac{7}{3} = \frac{23}{3} \text{ square units.}$$

Remark: the naive single integral gives $\int_0^3 (x^2 - 4)dx = -3$, which is not the area — a classic GCE trap.

10.2 Area between Two Curves

METHOD **Area between $y = f(x)$ and $y = g(x)$**

1. Find the intersection points: solve $f(x) = g(x)$; call the solutions $a < b$ (these are usually the limits).
2. On $[a, b]$, identify the upper curve f and the lower curve g (test one value).
3. Compute $A = \int_a^b [f(x) - g(x)]\,dx$.
4. If the curves cross inside the interval, split and take absolute values of each piece.

Reflex: the formula $A = \int_a^b (\text{top} - \text{bottom})\,dx$ works even if both curves are below the x-axis.

WORKED EXAMPLE 26 **Area between $y = x^2$ and $y = 2x$**

Find the area of the finite region enclosed by the parabola $y = x^2$ and the line $y = 2x$.

SOLUTION

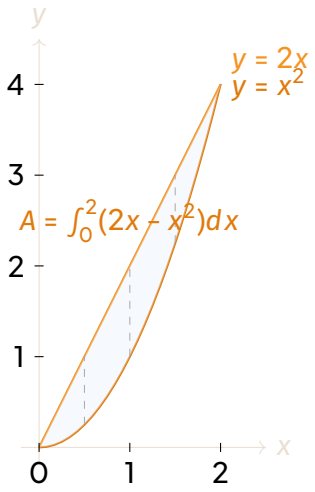
Step 1: intersections. $x^2 = 2x \iff x(x - 2) = 0 \iff x = 0$ or $x = 2$. The limits are $a = 0$, $b = 2$.
Step 2: which curve is on top? Test $x = 1$: line gives $y = 2$, parabola gives $y = 1$. The line $y = 2x$ is above on $(0, 2)$.
Step 3: integrate top minus bottom.

$$A = \int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2$$
$$= \left(4 - \frac{8}{3}\right) - 0 = \frac{12 - 8}{3} = \frac{4}{3}.$$

The area of the region is $\frac{4}{3}$ square units.
Check with a sketch: the region is a small "lens" between the parabola and the chord joining $(0, 0)$ to $(2, 4)$; a value slightly bigger than 1 is plausible. ✓

EXPERIMENT **Visualising Area between Curves**

The shaded region below represents the area between $y = x^2$ (lower) and $y = 2x$ (upper) from $x = 0$ to $x = 2$.



Area between two curves: top minus bottom.

11 AREAS BOUNDED BY THE Y-AXIS

11.1 Integrating with Respect to y

METHOD **Integrating with respect to y**

For a region bounded by the curve $x = g(y)$, the y-axis, and the horizontal lines $y = c$, $y = d$:

1. Rewrite the equation of the curve as x in terms of y: $x = g(y)$.
2. Identify the horizontal bounds $y = c$ and $y = d$ (from given lines or intersection points).
3. Compute $A = \int_c^d g(y)\,dy$ if $g(y) \geq 0$; more generally $A = \int_c^d |g(y)|\,dy$, or $\int_c^d$(right curve − left curve)$\,dy$ between two curves.

Reflex: "bounded by the y-axis" ⇒ integrate with respect to y; the roles of x and y are swapped.

WORKED EXAMPLE 27 **Area between $y^2 = 4x$, the y-axis and $y = 4$**

Find the area of the region in the first quadrant bounded by the curve $y^2 = 4x$, the y-axis and the line $y = 4$.

SOLUTION

Step 1: express x as a function of y. From $y^2 = 4x$ we get $x = \dfrac{y^2}{4}$.
Step 2: bounds. The region runs from $y = 0$ (the x-axis, where the curve meets the y-axis at the origin) to $y = 4$. For $y \in [0, 4]$, $x = \frac{y^2}{4} \geq 0$, so the curve is to the right of the y-axis.
Step 3: integrate with respect to y.

$$A = \int_0^4 \frac{y^2}{4}\,dy = \frac{1}{4}\left[\frac{y^3}{3}\right]_0^4 = \frac{1}{4}\cdot\frac{64}{3} = \frac{16}{3}.$$

The area is $\dfrac{16}{3}$ square units.
Alternative check (integrating in x): the curve is $y = 2\sqrt{x}$; at $y = 4$, $x = 4$. The rectangle $[0, 4] \times [0, 4]$ has area 16, and the area under the curve is $\int_0^4 2\sqrt{x}\,dx = \left[\frac{4}{3}x^{3/2}\right]_0^4 = \frac{4}{3}\cdot 8 = \frac{32}{3}$. Hence the region between the curve and the y-axis has area $16 - \frac{32}{3} = \frac{16}{3}$. ✓

12 VOLUMES OF SOLIDS OF REVOLUTION

12.1 Revolution about the x-axis: Disc and Washer Methods

METHOD **Disc method about the x-axis**

The region under $y = f(x) \geq 0$ from $x = a$ to $x = b$, rotated through 360° about the x-axis, generates a solid of volume

$$V = \pi \int_a^b [f(x)]^2 dx.$$

1. Sketch the region and confirm it touches the axis of rotation (otherwise use the washer method below).
2. Square the function — before integrating: $V = \pi \int y^2\,dx$, never $\pi\left(\int y\,dx\right)^2$.
3. Integrate and multiply by π; leave the answer in terms of π unless told otherwise.

METHOD **Washer method about the x-axis**

If the region is bounded by two curves $y = f(x)$ (outer) and $y = g(x)$ (inner) with $f(x) \geq g(x) \geq$

0 on $[a, b]$, rotation about the x-axis gives a hollow solid of volume

$$V = \pi \int_a^b \left[f(x)^2 - g(x)^2\right] dx.$$

WORKED EXAMPLE 28 **Volume generated by $y = \sqrt{x}$ and by a region between two curves**

(a) The region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$ is rotated through 360° about the x-axis. Find the volume generated.
(b) The region enclosed by $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume generated.

SOLUTION

(a) Apply the disc formula with $f(x) = \sqrt{x}$:

$$V = \pi \int_0^4 \left(\sqrt{x}\right)^2 dx = \pi \int_0^4 x\, dx = \pi \left[\frac{x^2}{2}\right]_0^4 = \pi \cdot \frac{16}{2} = 8\pi.$$

The volume is 8π cubic units.
(b) Step 1: intersections. $x = x^2 \iff x(1 - x) = 0 \iff x = 0$ or $x = 1$.
Step 2: upper and lower curves. On $(0, 1)$, test $x = \frac{1}{2}$: $y = x$ gives $\frac{1}{2}$, $y = x^2$ gives $\frac{1}{4}$. So $y = x$ is the outer radius, $y = x^2$ the inner radius.
Step 3: washer formula.

$$V = \pi \int_0^1 \left[x^2 - \left(x^2\right)^2\right] dx = \pi \int_0^1 \left(x^2 - x^4\right) dx$$
$$= \pi \left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5}\right) = \pi \cdot \frac{5 - 3}{15} = \frac{2\pi}{15}.$$

The volume is $\frac{2\pi}{15}$ cubic units.

12.2 Revolution about the y-axis: Disc Method

METHOD **Disc method about the y-axis**

The region between $x = g(y) \geq 0$ and the y-axis, from $y = c$ to $y = d$, rotated about the y-axis, gives

$$V = \pi \int_c^d \left[g(y)\right]^2 dy = \pi \int_c^d x^2\, dy.$$

1. Express x^2 in terms of y from the equation of the curve.

2. Convert the bounds to y-values.
3. Integrate x^2 with respect to y and multiply by π.

Reflex: rotation about the y-axis ⇒ integrate x^2 with respect to y; rotation about the x-axis ⇒ integrate y^2 with respect to x.

WORKED EXAMPLE 29 **Volume of a paraboloid**

The region bounded by the curve $y = x^2$, the line $y = 4$ and the y-axis (in the first quadrant) is rotated through 360° about the y-axis. Find the volume of the solid formed.

SOLUTION

Step 1: express x^2 in terms of y. From $y = x^2$ we have directly $x^2 = y$.
Step 2: bounds in y. The region extends from $y = 0$ (origin) to $y = 4$.
Step 3: apply the formula.

$$V = \pi \int_0^4 x^2 \, dy = \pi \int_0^4 y \, dy = \pi \left[\frac{y^2}{2} \right]_0^4 = \pi \cdot \frac{16}{2} = 8\pi.$$

The volume of the paraboloid is 8π cubic units.
Consistency check: the solid fits inside the cylinder of radius 2 and height 4 (volume $\pi r^2 h = 16\pi$), and 8π is exactly half of it — a classical result for a paraboloid. ✓

12.3 Alternative: Cylindrical Shell Method

METHOD **Shell method about the y-axis**

The region under $y = f(x) \geq 0$ from $x = a$ to $x = b$ ($0 \leq a < b$), rotated about the y-axis, can also be computed by cylindrical shells:

$$V = 2\pi \int_a^b x \, f(x) \, dx.$$

Each vertical strip at x of width dx generates a cylindrical shell of radius x, height $f(x)$, and thickness dx.

1. Identify the radius x and height $f(x)$ of a typical shell.
2. Integrate $2\pi x f(x)$ with respect to x from a to b.

Note: The shell method is often simpler when the disc/washer method would require solving for x in terms of y.

WORKED EXAMPLE 30 **Shell method: same paraboloid**

Use the shell method to find the volume of the solid generated by rotating the region bounded by $y = x^2$, $y = 4$, and the y-axis (first quadrant) about the y-axis.

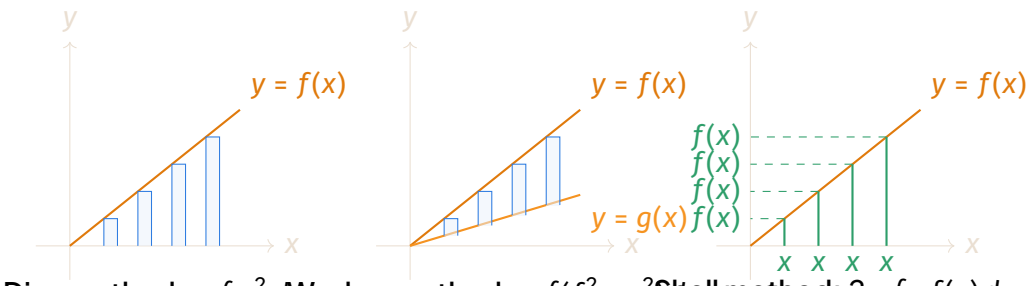
SOLUTION

The region is described by x from 0 to 2 (since $y = 4 \Rightarrow x = 2$). A vertical strip at x has height $4 - x^2$ (from the curve to the line $y = 4$). The shell volume is

$$V = 2\pi \int_0^2 x\,(4 - x^2)\,dx = 2\pi \int_0^2 (4x - x^3)\,dx = 2\pi \left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2\pi\,(8 - 4) = 8\pi.$$

Same result as the disc method. ✓

EXPERIMENT **Visualising Discs, Washers, and Shells**



Disc method: $\pi \int y^2\,dx$ Washer method: $\pi \int (f^2 - g^2)\,dx$ Shell method: $2\pi \int x f(x)\,dx$

Three methods for volumes of revolution.

13 CENTROIDS AND CENTRES OF MASS BY INTEGRATION

13.1 Centroid of a Plane Region

METHOD **Centroid of a region under a curve**

For the region under $y = f(x) \geq 0$ from $x = a$ to $x = b$, of area $A = \int_a^b f(x)\,dx$, the centroid $(\bar{x}, \bar{y})$

is given by

$$\bar{x} = \frac{1}{A}\int_a^b x\,f(x)\,dx, \qquad \bar{y} = \frac{1}{2A}\int_a^b \big[f(x)\big]^2 dx.$$

1. Compute the area A first.
2. Compute the two "moment" integrals $\int_a^b xf(x)\,dx$ and $\frac{1}{2}\int_a^b f(x)^2 dx$.
3. Divide by A; check that $(\bar{x}, \bar{y})$ lies inside (or on the boundary of) the region.

METHOD **Centroid of a region between two curves**

For the region bounded by $y = f(x)$ (upper) and $y = g(x)$ (lower) on $[a, b]$, with $f(x) \geq g(x)$, area $A = \int_a^b [f(x) - g(x)]\,dx$, the centroid is

$$\bar{x} = \frac{1}{A}\int_a^b x\big[f(x) - g(x)\big]\,dx, \qquad \bar{y} = \frac{1}{2A}\int_a^b \big[f(x)^2 - g(x)^2\big]\,dx.$$

WORKED EXAMPLE 31 **Centroid of the region under $y = 4 - x^2$**

Find the coordinates of the centroid of the region bounded by the parabola $y = 4 - x^2$ and the x-axis.

SOLUTION

Step 0: bounds. $4 - x^2 = 0 \iff x = \pm 2$. The region is symmetric about the y-axis, so we expect $\bar{x} = 0$; we verify by computation.
Step 1: area.

$$A = \int_{-2}^{2}(4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{32}{3}.$$

Step 2: moment about the y-axis.

$$\int_{-2}^{2} x(4 - x^2)\,dx = \int_{-2}^{2}(4x - x^3)\,dx.$$

The integrand is an odd function integrated over a symmetric interval, so the integral is 0.
Hence $\bar{x} = \frac{0}{A} = 0$, confirming symmetry.
Step 3: moment about the x-axis.

$$\int_{-2}^{2}(4 - x^2)^2\,dx = \int_{-2}^{2}\left(16 - 8x^2 + x^4\right)dx = \left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_{-2}^{2}$$
$$= 2\left(32 - \frac{64}{3} + \frac{32}{5}\right) = 2 \cdot \frac{480 - 320 + 96}{15} = 2 \cdot \frac{256}{15} = \frac{512}{15}.$$

Therefore

$$\bar{y} = \frac{1}{2A} \cdot \frac{512}{15} = \frac{1}{2} \cdot \frac{3}{32} \cdot \frac{512}{15} = \frac{3 \cdot 512}{2 \cdot 32 \cdot 15} = \frac{1536}{960} = \frac{8}{5}.$$

Conclusion: the centroid is $\left(0, \frac{8}{5}\right)$. Since $0 < \frac{8}{5} < 4$, the point lies inside the region. ✓

13.2 Centroid of a Solid of Revolution

METHOD **Centroid of a solid of revolution about the x-axis**

The solid generated by rotating the region under $y = f(x) \geq 0$ on $[a, b]$ about the x-axis has volume $V = \pi \int_a^b f(x)^2 \, dx$. By symmetry, the centroid lies on the x-axis: $\bar{y} = \bar{z} = 0$. The x-coordinate is

$$\bar{x} = \frac{1}{V} \int_a^b x \, \pi f(x)^2 \, dx = \frac{\int_a^b x f(x)^2 \, dx}{\int_a^b f(x)^2 \, dx}.$$

WORKED EXAMPLE 32 **Centroid of a paraboloid**

Find the centroid of the solid generated by rotating the region under $y = \sqrt{x}$ on $[0, 4]$ about the x-axis.

SOLUTION

Volume: $V = \pi \int_0^4 x \, dx = 8\pi$. Moment: $\int_0^4 x \cdot \pi x \, dx = \pi \int_0^4 x^2 \, dx = \pi \left[\frac{x^3}{3}\right]_0^4 = \frac{64\pi}{3}$. Thus $\bar{x} = \frac{64\pi/3}{8\pi} = \frac{8}{3}$. The centroid is $\left(\frac{8}{3}, 0, 0\right)$. ✓

13.3 Centre of Mass of a Rod with Variable Density

METHOD **Centre of mass of a non-uniform rod**

A rod of length L lies along the x-axis from $x = a$ to $x = b$ with linear density $\rho(x)$ (mass per unit length). Its total mass is $M = \int_a^b \rho(x) \, dx$, and the centre of mass is at

$$\bar{x} = \frac{1}{M} \int_a^b x \, \rho(x) \, dx.$$

WORKED EXAMPLE 33 **Rod with density proportional to distance**

A rod of length 2 m lies on the x-axis from $x = 0$ to $x = 2$. Its density is $\rho(x) = kx$ kg/m. Find the centre of mass.

SOLUTION

Mass: $M = \int_0^2 kx\,dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k$. Moment: $\int_0^2 x(kx)\,dx = k\int_0^2 x^2\,dx = k\left[\frac{x^3}{3}\right]_0^2 = \frac{8k}{3}$. Centre of mass: $\bar{x} = \frac{8k/3}{2k} = \frac{4}{3}$ m from the end $x = 0$. ✓

14 TRAPEZOID RULE AND THE MEAN VALUE THEOREM FOR INTEGRALS

14.1 Trapezoid Rule

METHOD **Trapezoid rule**

To approximate $\int_a^b f(x)\,dx$ with n strips of equal width $h = \frac{b-a}{n}$ and ordinates $y_0, y_1, \ldots, y_n$ at $x_k = a + kh$:

$$\int_a^b f(x)\,dx \approx \frac{h}{2}\left[y_0 + 2\left(y_1 + y_2 + \cdots + y_{n-1}\right) + y_n\right].$$

1. Compute h and list the ordinates in a table (work to at least 4 decimal places).
2. Apply the pattern: first + last, twice all the middle ones, times $\frac{h}{2}$.
3. State the answer to the accuracy requested.

WORKED EXAMPLE 34 **Trapezoid estimate**

Use the trapezoid rule with 4 strips to estimate $\int_0^2 \frac{1}{1+x^2}\,dx$, giving your answer to 3 decimal places.

SOLUTION

Here $a = 0$, $b = 2$, $n = 4$, so $h = \frac{2-0}{4} = 0.5$. Ordinates of $f(x) = \frac{1}{1+x^2}$:

x	0	0.5	1.0	1.5	2.0
$y = \frac{1}{1+x^2}$	1.0000	0.8000	0.5000	0.3077	0.2000

Apply the rule:

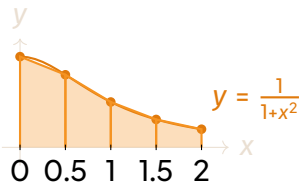
$$\int_0^2 \frac{dx}{1+x^2} \approx \frac{0.5}{2}\left[y_0 + 2(y_1 + y_2 + y_3) + y_4\right]$$
$$= 0.25\left[1.0000 + 2(0.8000 + 0.5000 + 0.3077) + 0.2000\right]$$
$$= 0.25\left[1.2000 + 2(1.6077)\right] = 0.25 \times 4.4154 = 1.10385.$$

So $\int_0^2 \frac{dx}{1+x^2} \approx 1.104$ (3 d.p.).

Comment: the exact value is $\left[\arctan x\right]_0^2 = \arctan 2 \approx 1.1071$; the trapezoid estimate with only 4 strips is already within 0.004.

EXPERIMENT **Visualising the Trapezoid Rule**

The figure shows the trapezoid approximation for $\int_0^2 \frac{dx}{1+x^2}$ with $n = 4$. The area under the curve is approximated by the sum of trapezoid areas.



Trapezoid rule with $n = 4$ strips.

14.2 Mean Value Theorem for Integrals

PROPERTY **Mean Value Theorem for Integrals**

If f is continuous on $[a, b]$, then there exists at least one number $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

The value $\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx$ is called the mean value (or average value) of f on $[a, b]$.

WORKED EXAMPLE 35 **Mean value of $\sin x$ on $[0, \pi]$**

Find the mean value of $f(x) = \sin x$ on $[0, \pi]$, and determine a value $c \in [0, \pi]$ at which $f(c)$ equals this mean value.

SOLUTION

The mean value of $f(x) = \sin x$ on $[0, \pi]$ is

$$\bar{f} = \frac{1}{\pi - 0} \int_0^{\pi} \sin x \, dx = \frac{1}{\pi}\left[- \cos x \right]_0^{\pi} = \frac{1}{\pi}(1 + 1) = \frac{2}{\pi}.$$

By the Mean Value Theorem for integrals (sin is continuous), there exists $c \in [0, \pi]$ with $\sin c = \frac{2}{\pi}$. Solving:

$$c = \arcsin\left(\frac{2}{\pi}\right) \approx \arcsin(0.6366) \approx 0.6901 \text{ rad},$$

or, by symmetry of the sine curve, $c = \pi - 0.6901 \approx 2.4515$ rad. Both values lie in $[0, \pi]$, illustrating the theorem.

WARNING **Frequent errors in this chapter**

- Forgetting the factor π or squaring after integrating in volume problems: it is $V = \pi \int y^2 \, dx$.
- Mixing the axes: about the x-axis use $y^2 \, dx$; about the y-axis use $x^2 \, dy$.
- Reporting a signed integral as an area when the curve crosses the axis.
- In the trapezoid rule, forgetting to double the interior ordinates or using h instead of $\frac{h}{2}$.
- In centroid problems, forgetting the factor $\frac{1}{2}$ in $\bar{y} = \frac{1}{2A} \int y^2 \, dx$.
- Confusing the formulas for centroid of a plane area and centroid of a solid of revolution.

15 EXERCISES

15.1 I check my knowledge

EXERCISE 7 MCQ: Riemann Sum Limit

Which of the following expressions correctly represents the definite integral $\int_0^1 (3x^2+1)\,dx$ as a limit of Riemann sums using a regular partition and right endpoints?

1. $\lim_{n\to\infty} \sum_{k=1}^{n} \left(3\left(\frac{k}{n}\right)^2+1\right)\frac{1}{n}$
2. $\lim_{n\to\infty} \sum_{k=1}^{n} \left(3\left(\frac{k-1}{n}\right)^2+1\right)\frac{1}{n}$
3. $\lim_{n\to\infty} \sum_{k=0}^{n-1} \left(3\left(\frac{k}{n}\right)^2+1\right)\frac{1}{n}$
4. $\lim_{n\to\infty} \sum_{k=1}^{n} \left(3k^2+1\right)\frac{1}{n}$

EXERCISE 8 True/False with Justification

Determine whether each statement is true or false. Justify your answer.

1. The area bounded by the curve $y = x^3$ and the x-axis between $x = -1$ and $x = 1$ is given by $\int_{-1}^{1} x^3\,dx$.
2. When calculating the area between two curves $y = f(x)$ and $y = g(x)$ on $[a, b]$ where $f(x) \geq g(x)$, the formula is $\int_a^b [f(x) - g(x)]\,dx$.
3. The volume of the solid generated by rotating the region under $y = f(x) \geq 0$ on $[a, b]$ about the y-axis is $V = \pi \int_a^b x^2 f(x)\,dx$.
4. The centroid $(\bar{x}, \bar{y})$ of a plane region of area A bounded by $y = f(x)$, $y = 0$, $x = a$, $x = b$ has $\bar{y} = \frac{1}{A}\int_a^b \frac{1}{2}[f(x)]^2\,dx$.

EXERCISE 9 Definitions and Direct Calculations

1. Write the definition of the definite integral $\int_a^b f(x)\,dx$ as the limit of a Riemann sum.
2. State the Mean Value Theorem for Integrals.
3. Evaluate directly: $\int_0^2 (4x - x^2)\,dx$.

4. Use the Trapezoid Rule with 4 strips (5 ordinates) to approximate $\int_0^2 x^2\,dx$.

EXERCISE 10 **Quick Applications**

1. Find the exact area enclosed by the curve $y = \sin x$ and the x-axis on the interval $[0, \pi]$.
2. The region bounded by $y = \sqrt{x}$, $x = 0$, $x = 4$ and the x-axis is rotated about the x-axis. Write down the integral for the volume (do not evaluate).
3. A uniform lamina occupies the region bounded by $y = x^2$ and $y = 4$. Write down the integral expressions for the coordinates of its centroid $(\bar{x}, \bar{y})$.
4. Given the velocity $v(t) = 3t^2 - 12t + 9$ m/s, find the average velocity over the interval $0 \le t \le 4$.

15.2 I practise

EXERCISE 11 **Area between Curve and x-axis**

Find the exact total area enclosed between the curve $y = x^2 - 2x$ and the x-axis. Sketch the curve, indicate the region, and justify carefully why the integral must be split.

EXERCISE 12 **Area between Two Curves**

Find the exact area of the finite region bounded by the curves $y = x^2$ and $y = 8\sqrt{x}$. Include a clear sketch showing the points of intersection and the representative strip.

EXERCISE 13 **Area with respect to y-axis**

Calculate the exact area of the region bounded by the curve $y^2 = 4x$, the y-axis, and the lines $y = 0$ and $y = 4$, by integrating with respect to y. Provide a sketch of the region.

EXERCISE 14 **Volume of Revolution about x-axis**

The region under the curve $y = \sqrt{\sin x}$ on the interval $[0, \pi]$ is rotated completely about the x-axis. Find the exact volume of the solid generated.

15.3 I prepare for the GCE

EXERCISE 15 **Volume by Washer Method – 8 marks**

The region R is bounded by the curves $y = x^2$ and $y = x$.

1. Sketch the region R, showing clearly the coordinates of the points of intersection. [2]
2. The region R is rotated through 360° about the x-axis. Using the washer method, show that the volume V of the solid generated is given by $V = \pi \int_0^1 (x^2 - x^4)\,dx$. [3]
3. Evaluate this integral to find the exact volume. [2]
4. If the same region R is rotated about the y-axis, write down (but do not evaluate) the integral for the volume using the shell method. [1]

EXERCISE 16 **Centroid and Centre of Mass – 10 marks**

1. Find the coordinates of the centroid of the plane region bounded by the parabola $y = 4 - x^2$ and the x-axis. [6]
2. A uniform solid of revolution is formed by rotating the curve $y = \sqrt{x}$, for $0 \le x \le 4$, about the x-axis. Determine the x-coordinate of the centre of mass of this solid. [4]

EXERCISE 17 **Trapezoid Rule and Mean Value Theorem – 9 marks**

1. Using the trapezoid rule with 5 ordinates, estimate the value of the integral $\int_0^1 \frac{1}{1+x^2}\,dx$. Give your answer correct to 4 decimal places. [4]
2. Hence deduce an approximation for π. [1]
3. The temperature T (in degrees Celsius) in Yaoundé over a 12-hour period is modelled by the function $T(t) = 22 + 5\sin\left(\frac{\pi t}{12}\right)$, where t is the time in hours after 6:00 am ($0 \le t \le 12$).
   (a) Find the mean temperature over this 12-hour period. [2]
   (b) State the Mean Value Theorem for Integrals and explain why it guarantees that there is at least one instant $t = c$ in $[0, 12]$ when the actual temperature equals this mean temperature. [2]

16 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: Riemann Sum Limit**

The correct answer is (a).
Justification. For a regular partition of $[0,1]$ into n subintervals, the width of each strip is $\Delta x = \frac{1-0}{n} = \frac{1}{n}$. The right endpoint of the k-th subinterval is $x_k = a + k\Delta x = \frac{k}{n}$, for $k = 1, 2, \dots, n$. Hence

$$\int_0^1 (3x^2 + 1)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k)\,\Delta x = \lim_{n\to\infty} \sum_{k=1}^{n} \left(3\left(\frac{k}{n}\right)^2 + 1\right)\frac{1}{n}.$$

Why the others fail:

- (b) and (c) use left endpoints $x_{k-1} = \frac{k-1}{n}$ (both describe the same left-endpoint sum);
- (d) evaluates f at k instead of $x_k = k/n$, so it does not even converge to a finite integral (the sum grows like n^2).

Note that (b) and (c) would still converge to the same integral (since f is continuous, any choice of sample point works), but the question specifically demands right endpoints, so only (a) is correct.

SOLUTION **Exercise 2 — True/False with Justification**

1. False. On $[-1, 0]$, $x^3 \le 0$, so $\int_{-1}^{1} x^3\,dx = 0$ (the function is odd) — this is the signed area, not the geometric area. The true area is

$$A = \int_{-1}^{1} |x^3|\,dx = -\int_{-1}^{0} x^3\,dx + \int_{0}^{1} x^3\,dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

2. True. If $f(x) \ge g(x)$ on $[a, b]$, the height of a vertical strip is $f(x) - g(x) \ge 0$, and summing gives $A = \int_a^b [f(x) - g(x)]\,dx$. This holds even if the curves cross the x-axis, since shifting both curves up by a constant does not change the area.

3. False. The formula given mixes methods incorrectly. Rotation about the y-axis by the shell method gives $V = 2\pi \int_a^b x\,f(x)\,dx$ (for $0 \le a < b$); by the disc/washer method one integrates with respect to y: $V = \pi \int_c^d [x(y)]^2\,dy$ (with washers if the region does not touch the axis). The factor 2π is missing and $x^2 f(x)$ has the wrong structure.

4. True. A vertical strip at position x has height $f(x)$, area $f(x)\,dx$, and its own centroid at height $\frac{1}{2}f(x)$. Taking moments about the x-axis:

$$A\bar{y} = \int_a^b \tfrac{1}{2} f(x) \cdot f(x)\,dx = \int_a^b \tfrac{1}{2}[f(x)]^2\,dx,$$

so $\bar{y} = \dfrac{1}{A}\displaystyle\int_a^b \tfrac{1}{2}[f(x)]^2\,dx$. □

SOLUTION **Exercise 3 — Definitions and Direct Calculations**

1. Partition $[a, b]$ into n subintervals $[x_{k-1}, x_k]$ of width $\Delta x_k = x_k - x_{k-1}$, and choose any sample point $\xi_k \in [x_{k-1}, x_k]$. Then

$$\int_a^b f(x)\,dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} f(\xi_k)\,\Delta x_k,$$

provided this limit exists and is independent of the partition and of the choice of ξ_k.

For a regular partition with right endpoints: $\displaystyle\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} f\!\left(a + k\frac{b-a}{n}\right)$.

2. Mean Value Theorem for Integrals. If f is continuous on $[a, b]$, then there exists at least one $c \in [a, b]$ such that

$$\int_a^b f(x)\,dx = f(c)\,(b-a), \qquad \text{i.e.} \qquad f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

3.

$$\int_0^2 (4x - x^2)\,dx = \left[2x^2 - \frac{x^3}{3}\right]_0^2 = \left(8 - \frac{8}{3}\right) - 0 = \frac{16}{3}.$$

4. Here $h = \dfrac{2-0}{4} = 0.5$, ordinates at $x = 0, 0.5, 1, 1.5, 2$ with $f(x) = x^2$:

x	0	0.5	1	1.5	2
$f(x)$	0	0.25	1	2.25	4

$$T = \frac{h}{2}\left[y_0 + y_4 + 2(y_1 + y_2 + y_3)\right] = \frac{0.5}{2}\left[0 + 4 + 2(0.25 + 1 + 2.25)\right] = 0.25 \times 11 = 2.75.$$

(The exact value is $\frac{8}{3} \approx 2.667$; the trapezoid rule overestimates here because x^2 is convex.)

SOLUTION **Exercise 4 — Quick Applications**

1. Since $\sin x \geq 0$ on $[0, \pi]$:

$$A = \int_0^{\pi} \sin x \, dx = [-\cos x]_0^{\pi} = -(-1) - (-1) = 2 \text{ square units.}$$

2. Disc method about the x-axis:

$$V = \pi \int_0^4 \left(\sqrt{x}\right)^2 dx = \pi \int_0^4 x \, dx.$$

3. The curves meet where $x^2 = 4$, i.e. $x = \pm 2$. By symmetry $\bar{x} = 0$. With $A = \int_{-2}^{2} (4 - x^2)\, dx$:

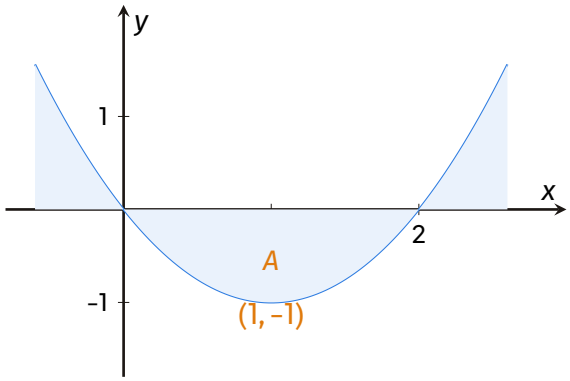
$$\bar{x} = \frac{1}{A} \int_{-2}^{2} x\,(4 - x^2)\, dx = 0, \qquad \bar{y} = \frac{1}{A} \int_{-2}^{2} \frac{1}{2}\left(4^2 - (x^2)^2\right) dx = \frac{1}{2A} \int_{-2}^{2} \left(16 - x^4\right) dx.$$

4. Average velocity = mean value of v on $[0, 4]$:

$$\bar{v} = \frac{1}{4 - 0} \int_0^4 (3t^2 - 12t + 9)\, dt = \frac{1}{4}\left[t^3 - 6t^2 + 9t\right]_0^4 = \frac{1}{4}(64 - 96 + 36) = \frac{4}{4} = 1 \,\text{m s}^{-1}.$$

SOLUTION **Exercise 5 — Area between Curve and x-axis**

Sketch and sign analysis. $y = x^2 - 2x = x(x - 2)$ is a parabola opening upwards, crossing the x-axis at $x = 0$ and $x = 2$, with vertex $(1, -1)$.



The region between $y = x^2 - 2x$ and the x-axis lies below the axis on $(0, 2)$.

Why split? On $[0, 2]$, $y = x(x - 2) \leq 0$, so $\int_0^2 (x^2 - 2x)\, dx$ is negative: it gives the signed area.

The geometric area requires $|y|$:

$$A = \int_0^2 |x^2 - 2x|\,dx = -\int_0^2 (x^2 - 2x)\,dx.$$

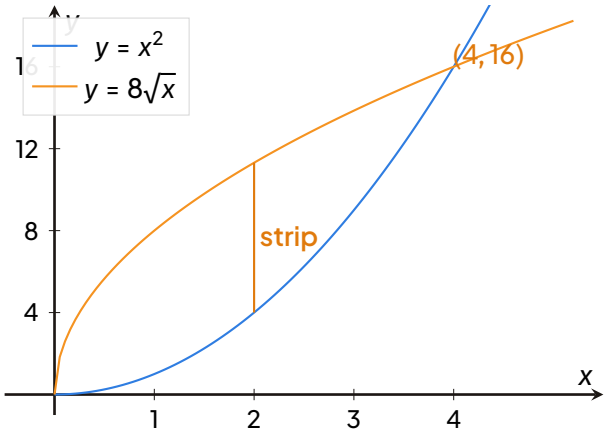
(Here a single piece suffices since the curve does not change sign inside $(0, 2)$; in general one splits at every root.)
Computation.

$$\int_0^2 (x^2 - 2x)\,dx = \left[\frac{x^3}{3} - x^2\right]_0^2 = \frac{8}{3} - 4 = -\frac{4}{3}.$$

$$\boxed{A = \frac{4}{3} \text{ square units}}$$

SOLUTION **Exercise 6 — Area between Two Curves**

Intersections. $x^2 = 8\sqrt{x}$ with $x \geq 0$: squaring, $x^4 = 64x$, so $x(x^3 - 64) = 0$, giving $x = 0$ or $x = 4$. Points: $(0, 0)$ and $(4, 16)$.
Which curve is above? Test $x = 1$: $x^2 = 1$ and $8\sqrt{x} = 8$. Hence $8\sqrt{x} \geq x^2$ on $[0, 4]$.



Region bounded by $y = x^2$ and $y = 8\sqrt{x}$; representative strip of height $8\sqrt{x} - x^2$.

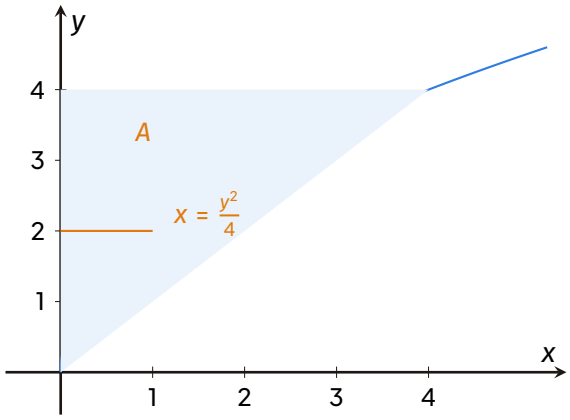
Area.

$$A = \int_0^4 \left(8\sqrt{x} - x^2\right)dx = \int_0^4 \left(8x^{1/2} - x^2\right)dx = \left[\frac{16}{3}x^{3/2} - \frac{x^3}{3}\right]_0^4$$

$$= \frac{16}{3}(8) - \frac{64}{3} = \frac{128 - 64}{3} = \boxed{\frac{64}{3} \text{ square units}}.$$

SOLUTION **Exercise 7 — Area with respect to y-axis**

Express x as a function of y. From $y^2 = 4x$ we get $x = \dfrac{y^2}{4}$ (right-hand parabola, $x \geq 0$). The region is bounded by this curve, the y-axis ($x = 0$), and $y = 0$, $y = 4$.



Horizontal strip of length $x = y^2/4$ and thickness dy.

Area. A horizontal strip at height y has length $\dfrac{y^2}{4} - 0$ and thickness dy:

$$A = \int_0^4 \frac{y^2}{4}\,dy = \frac{1}{4}\left[\frac{y^3}{3}\right]_0^4 = \frac{1}{4}\cdot\frac{64}{3} = \boxed{\frac{16}{3}\text{ square units}}.$$

SOLUTION **Exercise 8 — Volume of Revolution about x-axis**

By the disc method, $V = \pi\displaystyle\int_0^{\pi} y^2\,dx$ with $y^2 = \sin x$:

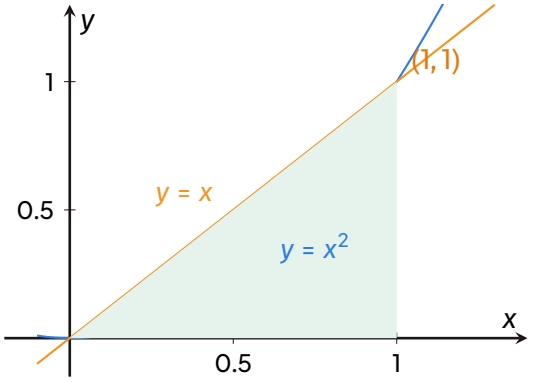
$$V = \pi\int_0^{\pi} \sin x\,dx = \pi\,[-\cos x]_0^{\pi} = \pi\,(-(-1) + 1) = \boxed{2\pi\text{ cubic units}}.$$

Remark. No splitting is needed: $\sin x \geq 0$ on $[0, \pi]$, so $y = \sqrt{\sin x}$ is real and non-negative throughout, and squaring removes the root exactly.

SOLUTION **Exercise 9 — Volume by Washer Method (GCE-type, 8 marks)**

1. Sketch [2 marks]. Intersections: $x^2 = x \Rightarrow x(x - 1) = 0 \Rightarrow x = 0, 1$. Points $(0, 0)$ and $(1, 1)$.

On $[0,1]$, $x \geq x^2$ (test $x = \frac{1}{2}$: $\frac{1}{2} > \frac{1}{4}$).



Region R between $y = x$ (outer radius) and $y = x^2$ (inner radius).

Examiner expects: both curves drawn, intersection coordinates labelled, region R shaded. [B1 curves, B1 points + shading]

2. Washer set-up [3 marks]. Rotating a vertical strip about the x-axis gives a washer (annulus) with outer radius $R(x) = x$ and inner radius $r(x) = x^2$. The cross-sectional area is

$$A(x) = \pi R^2 - \pi r^2 = \pi\left(x^2 - x^4\right).$$

Summing (integrating) these slices from $x = 0$ to $x = 1$:

$$V = \int_0^1 A(x)\,dx = \pi \int_0^1 \left(x^2 - x^4\right) dx. \qquad \text{[M1 washer area, A1 limits, A1 result]}$$

3. Evaluation [2 marks].

$$V = \pi\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \pi\left(\frac{1}{3} - \frac{1}{5}\right) = \boxed{\frac{2\pi}{15} \text{ cubic units}}. \qquad \text{[M1 integration, A1 exact value]}$$

4. Shell method about the y-axis [1 mark]. A vertical strip at x has height $x - x^2$ and rotates into a cylindrical shell of radius x:

$$V = 2\pi \int_0^1 x\left(x - x^2\right) dx. \qquad \text{[B1]}$$

Typical pitfalls penalised: writing $\pi \int (x - x^2)^2 dx$ (squaring the difference instead of subtracting the squares), and forgetting the factor 2π in the shell formula.

SOLUTION **Exercise 10 — Centroid and Centre of Mass (GCE-type, 10 marks)**

1. Centroid of the region under $y = 4 - x^2$ [6 marks].

   Zeros: $4 - x^2 = 0 \Rightarrow x = \pm 2$. The curve is above the x-axis on $[-2, 2]$.

   Area:

$$A = \int_{-2}^{2} (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{32}{3}. \quad \text{[M1 A1]}$$

   $\bar{x}$: The region is symmetric about the y-axis (equivalently, $x(4-x^2)$ is odd, so its integral over $[-2, 2]$ vanishes):

$$\bar{x} = \frac{1}{A}\int_{-2}^{2} x(4 - x^2)\,dx = 0. \quad \text{[M1 A1]}$$

   $\bar{y}$: Using $\bar{y} = \dfrac{1}{A}\displaystyle\int_{-2}^{2} \tfrac{1}{2}y^2\,dx$:

$$\int_{-2}^{2} \tfrac{1}{2}(4 - x^2)^2\,dx = \frac{1}{2}\int_{-2}^{2} (16 - 8x^2 + x^4)\,dx = \frac{1}{2}\left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_{-2}^{2}.$$

   At $x = 2$: $32 - \dfrac{64}{3} + \dfrac{32}{5} = \dfrac{480 - 320 + 96}{15} = \dfrac{256}{15}$; doubling (even function) and halving:

$$\frac{1}{2} \cdot 2 \cdot \frac{256}{15} = \frac{256}{15}.$$

   Hence

$$\bar{y} = \frac{256/15}{32/3} = \frac{256}{15} \times \frac{3}{32} = \frac{8}{5}. \quad \text{[M1 A1]}$$

$$\boxed{(\bar{x}, \bar{y}) = \left(0, \tfrac{8}{5}\right)}$$

2. Centre of mass of the solid from $y = \sqrt{x}$, $0 \le x \le 4$ [4 marks].

   By symmetry the centre of mass lies on the x-axis. For a uniform solid of revolution, using discs of cross-section $\pi y^2 = \pi x$:

$$\bar{x} = \frac{\displaystyle\int_0^4 x \cdot \pi y^2\,dx}{\displaystyle\int_0^4 \pi y^2\,dx} = \frac{\displaystyle\int_0^4 x^2\,dx}{\displaystyle\int_0^4 x\,dx}. \quad \text{[M1]}$$

$$\int_0^4 x^2\,dx = \left[\frac{x^3}{3}\right]_0^4 = \frac{64}{3}, \qquad \int_0^4 x\,dx = \left[\frac{x^2}{2}\right]_0^4 = 8. \quad \text{[M1 A1]}$$

$$\bar{x} = \frac{64/3}{8} = \boxed{\frac{8}{3}}. \quad \text{[A1]}$$

Examiner expectations: correct use of the disc moment formula $\int x\,\pi y^2 dx$ in the numerator; the density ρ and π cancel, which should be noted. A common error is weighting by y instead of by the cross-sectional area πy^2.

SOLUTION **Exercise 11 — Trapezoid Rule and Mean Value Theorem (GCE-type, 9 marks)**

1. Trapezoid rule, 5 ordinates [4 marks]. Five ordinates means 4 strips: $h = \frac{1-0}{4} = 0.25$, with $f(x) = \frac{1}{1+x^2}$:

x	0	0.25	0.5	0.75	1
$f(x)$	1	0.941176	0.8	0.64	0.5

[B1 correct h and ordinates, M1 formula]

$$T = \frac{h}{2}\left[y_0 + y_4 + 2(y_1 + y_2 + y_3)\right]$$
$$= \frac{0.25}{2}\left[1 + 0.5 + 2(0.941176 + 0.8 + 0.64)\right]$$
$$= 0.125 \times [1.5 + 2(2.381176)] = 0.125 \times 6.262353 = 0.782794.$$

$$\boxed{\int_0^1 \frac{dx}{1+x^2} \approx 0.7828 \text{ (4 d.p.)}}$$ [A1 value, A1 rounding]

2. Approximation of π [1 mark]. Since $\int_0^1 \frac{dx}{1+x^2} = [\arctan x]_0^1 = \frac{\pi}{4}$, we get

$$\pi \approx 4 \times 0.7828 = \boxed{3.1312}.$$ [B1]

3. (a) Mean temperature [2 marks].

$$\bar{T} = \frac{1}{12}\int_0^{12}\left(22 + 5\sin\frac{\pi t}{12}\right)dt = \frac{1}{12}\left[22t - \frac{60}{\pi}\cos\frac{\pi t}{12}\right]_0^{12}.$$ [M1]

$$= \frac{1}{12}\left[\left(264 - \frac{60}{\pi}\cos\pi\right) - \left(0 - \frac{60}{\pi}\cos 0\right)\right] = \frac{1}{12}\left(264 + \frac{120}{\pi}\right) = 22 + \frac{10}{\pi}.$$

$$\boxed{\bar{T} = 22 + \frac{10}{\pi} \approx 25.18^\circ\text{C}}$$ [A1]

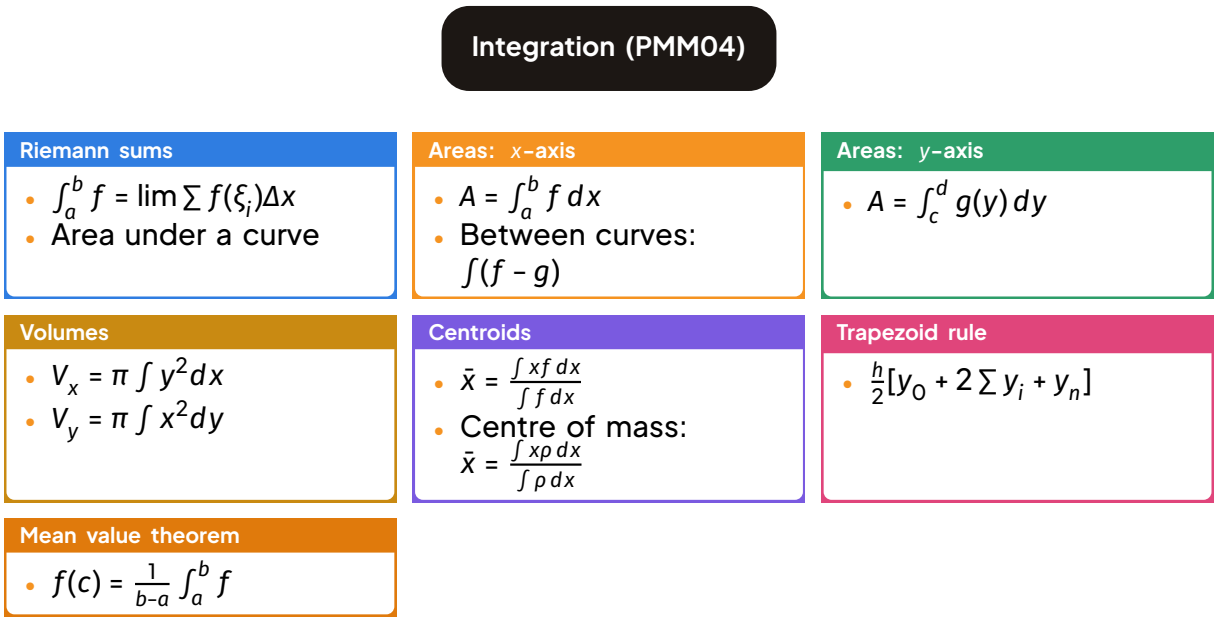
(b) MVT for Integrals [2 marks]. Statement: if f is continuous on $[a, b]$, there exists $c \in [a, b]$ such that $\int_a^b f(x)\,dx = f(c)(b-a)$. [B1]

Here $T(t) = 22+5\sin\left(\frac{\pi t}{12}\right)$ is continuous on $[0, 12]$ (a sum of continuous functions). The MVT therefore guarantees some $c \in [0, 12]$ with $T(c) = \bar{T} = 22 + \frac{10}{\pi}$: at that instant the actual temperature equals the 12-hour mean temperature. [B1]

(Indeed $\sin\frac{\pi c}{12} = \frac{2}{\pi}$ gives $c = \frac{12}{\pi}\arcsin\frac{2}{\pi} \approx 2.63$ h, i.e. about 8:38 am — existence is what the theorem guarantees.)

Examiner expectations: in (a), the formula $\frac{h}{2}[y_0 + y_n + 2(\text{middle ordinates})]$ must be quoted or clearly used; premature rounding of ordinates below 5 d.p. loses the final accuracy mark. In (c)(ii), continuity of T must be explicitly invoked — the MVT does not apply to discontinuous functions.

17 SUMMARY SHEET



Mind map of the chapter Integration: from Riemann sums to applications.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions.

- A **Riemann sum** of f on $[a, b]$ is $\sum_{i=1}^{n} f(\xi_i)\,\Delta x_i$ where $\Delta x_i = x_i - x_{i-1}$ and $\xi_i \in [x_{i-1}, x_i]$.
- The **definite integral** is the limit of Riemann sums as the mesh tends to 0: $\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(\xi_i)\,\frac{b-a}{n}$ (for equal subdivisions).
- $\int_a^b f(x)\,dx = F(b) - F(a)$ where F is any primitive of f (Fundamental Theorem).

Formulas to know by heart.

- Area under a curve ($f \geq 0$ on $[a, b]$): $A = \int_a^b f(x)\,dx$. If $f \leq 0$, $A = -\int_a^b f(x)\,dx$ (area is always positive).
- Area between two curves ($f \geq g$ on $[a, b]$): $A = \int_a^b [f(x) - g(x)]dx$.
- Area bounded by the y-axis ($x = g(y) \geq 0$): $A = \int_c^d g(y)\,dy$.

- Volume of revolution about the x-axis: $V = \pi \int_a^b [f(x)]^2 \, dx$.
- Volume of revolution about the y-axis: $V = \pi \int_c^d [g(y)]^2 \, dy$.
- Volume of a washer (region between f and g rotated about the x-axis): $V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2\right) dx$.
- Centroid of the region under $y = f(x)$, $a \le x \le b$: $\bar{x} = \dfrac{\int_a^b x f(x) \, dx}{\int_a^b f(x) \, dx}$, $\bar{y} = \dfrac{\frac{1}{2}\int_a^b [f(x)]^2 dx}{\int_a^b f(x) \, dx}$.
- Centre of mass of a rod of density $\rho(x)$: $\bar{x} = \dfrac{\int_a^b x \, \rho(x) \, dx}{\int_a^b \rho(x) \, dx}$; for a lamina use moments $M_y = \int x \, dA$, $M_x = \int y \, dA$.
- **Trapezoid rule** with n strips of width $h = \dfrac{b-a}{n}$: $\int_a^b f(x) \, dx \approx \dfrac{h}{2}\left[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\right]$.
- **Mean value theorem for integrals**: if f is continuous on $[a, b]$, there exists $c \in [a, b]$ with $f(c) = \dfrac{1}{b-a}\int_a^b f(x) \, dx$ (mean value of f).

Methods.

- To compute an area: sketch the curve(s), find intersections (solve $f(x) = g(x)$), decide on the variable of integration, split the integral where the sign or the upper curve changes, then integrate.
- To compute a volume: identify the axis of rotation, express the radius as a function of the variable of integration, square it, multiply by π, integrate.
- Trapezoid rule: build the table of ordinates $y_0, \ldots, y_n$ at $x_i = a + ih$, then apply the formula; more strips give a better approximation.

Common mistakes.

- Forgetting that an integral can be negative while an area is always positive: take absolute values or split at the zeros of f.
- Forgetting the factor π or forgetting to square the radius in volume formulas.
- Mixing the axes: $\pi \int y^2 dx$ is about the x-axis; $\pi \int x^2 dy$ is about the y-axis.
- In the trapezoid rule, forgetting the factor 2 on interior ordinates or using h instead of $h/2$.
- Wrong limits when integrating with respect to y: limits must be y-values.

GCE tips.

- Always sketch the region first: most marks are lost through wrong limits, not wrong integration.
- State the formula before substituting; examiners award method marks.
- Give volumes and areas as exact values (in terms of π) unless a decimal is requested.
- For the trapezoid rule, keep at least 4 decimal places in intermediate ordinates.
- Check plausibility: compare a volume with that of the containing cylinder, an area with that of a bounding rectangle.

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